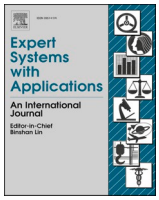
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Enhanced goal direction and remodeling of population distributions multimodal multi-objective evolutionary algorithm

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ABSTRACT

In recent years, multimodal multi-objective optimization problems (MMOPs) have become a prominent research focus in the field of computational evolution, increasing the demands on the performance of multimodal multi-objective evolutionary algorithms (MMOEAs). To evaluate the merits of MMOEA, it is usually necessary to satisfy the following three key criteria: (1) good convergence, (2) the ability to find more equivalent Pareto solutions (PSs), and (3) a uniform population distribution in the decision space and objective space. However, most current algorithms fail to satisfy all the above criteria simultaneously when facing challenges such as search tasks of varying difficulty and uneven allocation of computational resources. To address these challenges, this paper proposes an enhanced goal direction and remodeling of population distributions multimodal multi-objective evolutionary algorithm (MMOEA-EGR). The algorithm dynamically selects evolutionary stages through reinforcement learning, flexibly guiding the population to evolve under the guidance of the objectives of each stage, thus promoting efficient collaboration among the stages in the whole optimization process. Meanwhile, the algorithm adopts a remodeling population distribution strategy to enhance the evolutionary efficiency while optimizing the diversity of the decision space. Experimental results show that MMOEA-EGR outperforms several mainstream multimodal multi-objective evolutionary algorithms on several MMOPs standard test sets.

1. Introduction

Multi-objective optimization problems (Nematollahi et al., 2018; Liu et al., 2019) (MOPs) require simultaneous optimization of multiple objective functions with competing relationships (Zhou et al., 2011). Finding a single globally optimal solution faces significant challenges due to the inherent contradictions among objective functions. Essentially, enhancing one objective often leads to degradation of the performance of other objectives. Therefore, research in this area focuses on harmonizing the trade-offs among multiple objectives to produce solution sets with trade-off properties. The mathematical model can be formulated as follows:

$$\begin{aligned} &Min\ F(x) = \{f_1(x),\ f_2(x), \ldots, f_m(x)\} \\ &s.t.\ x \in\ \Omega \end{aligned} \tag{1}$$

where x is the decision variable, m is the number of objectives to be optimized, and Ω denotes the feasible solution domain. In this solution domain, for any two solutions x_1 and x_2, x_2 is said to dominate x_1 if x_2 is noninferior to x_1 on all objectives and strictly superior to x_1 on at least one metric (denoted as $x_1 \prec x_2$). A solution is said to be Pareto-optimal when it is not dominated by any other solution in the decision space and reaches Pareto optimality. The set of all such Pareto optimal solutions is called the Pareto optimal solution set (PS) (Figueiredo et al., 2016). The front formed by the mapping of these optimal solutions in the objective space is the Pareto front (PF) (Zapotecas-Martínez et al., 2023). Many multi-objective optimization algorithms (MOEAs) have been proposed and have shown excellent performance (Deb et al., 2002; Zhang & Li, 2007; Zitzler et al., 2001).

Real-world multi-objective optimization problems often involve multiple equivalent Pareto optimal sets (Hu et al., 2022), known as multimodal multi-objective optimization problems (MMOPs). For example, Fig. 1 shows two solutions (A and B) in the decision space, belonging to distinct Pareto sets (PS1 and PS2), yet both map to the same point M in the objective space. MMOPs are very common in reality, such as the map path planning problem (Yue et al., 2017), raw material purchasing problem (Zhang et al., 2019), multi-objective knapsack problem (Jaszkiewicz, 2002), feature selection problem (Yue et al.,

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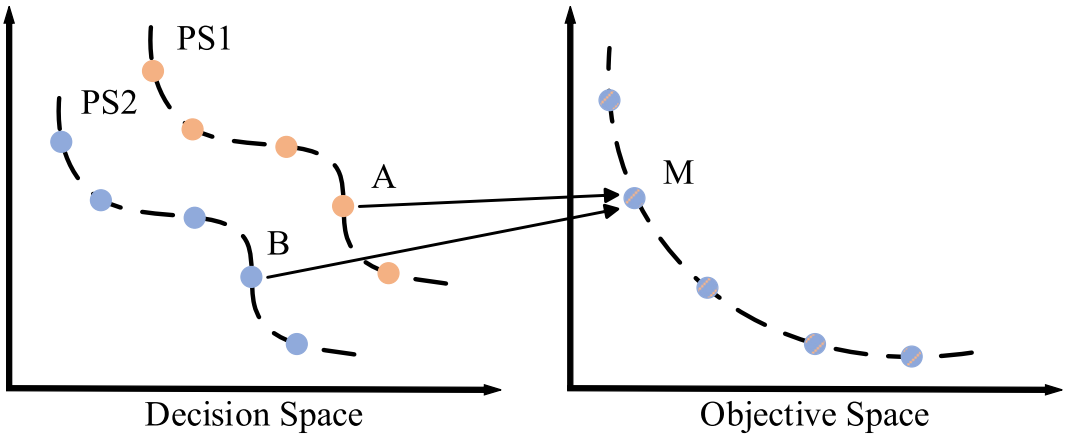


Fig. 1. Schematic diagram of a multimodal multi-objective optimization problem.

2019), and protein design problem (Lapizco-Encinas et al., 2010) etc.

Recently, multimodal multi-objective optimization problems have garnered significant attention in the field of evolutionary computation. Effectively solving such problems requires not only finding solutions that approximate the true Pareto front, but also being able to uncover and distinguish all equivalent Pareto solution sets. This places high demands on the algorithm's ability to search in the decision space. Classical MOEAs commonly face the problem of population diversity decline triggered by premature convergence when responding to MMOPs. This phenomenon causes the algorithm search process to often fall into a localized region of the decision space, making it difficult to effectively detect other solution set regions with equivalent Pareto frontier characteristics. To address this typical challenge, several innovative improvement strategies have been developed in the academic community, including representative approaches such as optimization strategies based on objective space relaxation (Ming et al., 2022) and dynamic parameter regulation mechanisms (Wang et al., 2020).

Zhang et al. (2020) proposed an adaptive niche maintenance mechanism inspired by isotropic magnetic repulsion to balance population distribution in the decision space. This approach dynamically guides solutions to spread across equivalent Pareto regions, improving decision-space diversity. Cao et al. (2024) designed a fitness function based on the niche method to assess individual diversity, used to calculate the fitness of solutions in the diversity archive, therefore improving the diversity of solutions in the decision space. Other studies employing niche techniques include (Hu, Zhou, Su, & Liu, 2022; Lin et al., 2020; Yang, Wang, Luo, & He, 2021; Zhou, Hu, Zhou, & Yuan, 2021; Zou, Deng, Zheng, & Yang, 2020). Wang et al. (2022) developed a decision variable pre-selection scheme that preserves more optimal solutions, further enhancing diversity. Hu et al. (2022) combined Particle Swarm Optimization (PSO) and Differential Evolution (DE) algorithms to propose two generation mechanisms with a dual-archive model, aiming to seek high-quality Pareto fronts in multi-objective optimization. Feng et al. (2024) introduced an improved crowding distance (MCD) in the MOPSO_MCD algorithm, enabling a more reasonable assessment of individual crowding, thus enhancing population diversity. Javadi et al. (2019) proposed the NSGA-II-WSCD-NBM algorithm, which is structurally similar to DN_NSGA-II but distinguishes the quality of solutions through a refined crowding distance strategy and neighborhood-based polynomial mutation strategy. Liang et al. (2020) proposed the MMODE_CSCD algorithm, which combines clustering and crowding distance techniques, and introduces a clustering-based special crowding distance to uniquely characterize each Pareto solution. Zhou et al. (2022) designed MMO_CDE, which also integrates clustering mechanisms and combines them with innovative mutation operators to significantly improve the algorithm's exploration ability.

The convergence of the objective space and the diversity of the decision space often form a mutually constraining relationship. In multimodal multi-objective evolutionary algorithms (MMOEA), maintaining diversity while enhancing search capability is particularly important (Liu et al., 2024). Ming et al. (2022) designed a co-evolutionary algorithm which optimizes the problem by using two populations working together. The dual-population framework operates through differentiated optimization mechanisms: the first subpopulation prioritizes convergence preservation and diversity maintenance, whereas the complementary subpopulation implements goal relaxation strategies to expand the explorative range within the decision space. This strategy helps to achieve a better balance between the breadth and depth of the search. Zhang et al. (2021) developed a dual-phase niche framework to augment algorithmic efficacy while enriching decision-space diversity. The initial phase employs a decision-space niche mechanism to drive comprehensive search-space exploration. Subsequently, the refined phase integrates dual-niche coordination with both decision-space solution distributions and objective-space solution attributes. In parallel, Xie et al. (2024) designed a fuzzy-preference evaluation metric for multidimensional fitness assessment of candidate solutions.

In summary, existing studies have mainly focused on fully assessing the exploration potential of individuals and improving the fitness of those individuals that are expected to discover more equivalent Pareto solutions to ensure their superiority in the evolutionary process so that they can be passed on to their offspring, thus enabling extensive exploration of the search space. However, this strategy still faces two major challenges:

1 Differences in search Difficulty: Due to the varying search difficulty of different equivalent Pareto sets, a large number of non-dominated solutions may be preferentially generated around some PSs, which may hinder the evolution of other potential solutions, thus increasing the difficulty of fully discovering PSs. In addition, different individuals have different needs for fitness enhancement, and if the fitness enhancement is not done in a proper way, it may limit the quality of the progeny individuals, which in turn affects the algorithm's effective exploration of other PSs.

2 Uneven distribution of computational resources: Even if the required fitness for potential solutions is identified and met during evolution, the scarcity of these solutions means that the available computational resources are limited. Multiple iterations may be needed to sufficiently explore their subspaces, reducing exploration efficiency.

In order to effectively address the above two challenges, in our previous study, we proposed MMOEA-GPD (Deng et al., 2024). Although MMOEA-GPD exhibits good performance in solving MMOPs, however, it still has several limitations. Specifically, the need to set the control parameters of each stage artificially leads to the insufficient generalizability of MMOEA-GPD when facing diverse MMOPs. In addition, its derivation metrics do not adequately trade off the convergence and

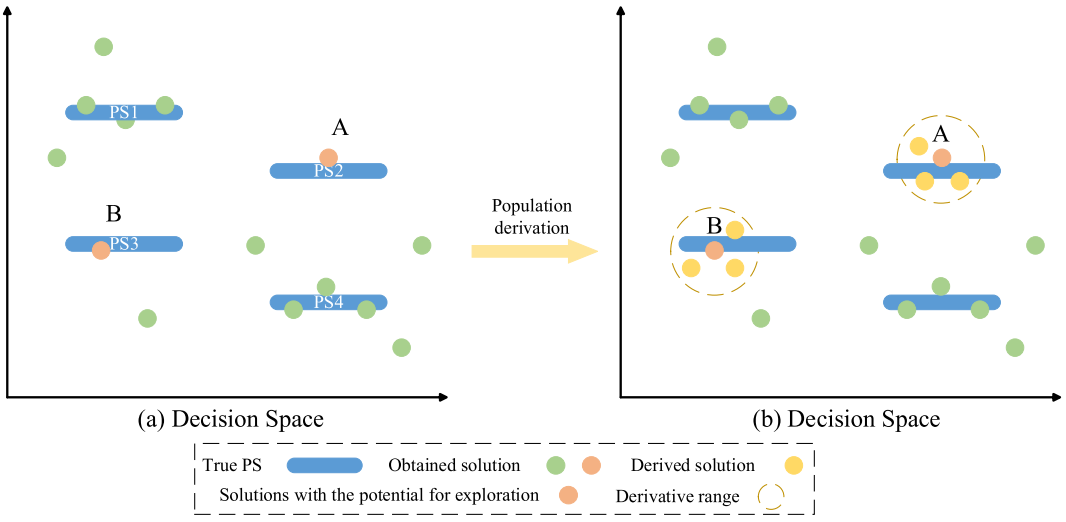


Fig. 2. Schematic diagram of the population derivation strategy.

diversity of individuals, and this shortcoming leads to the inability to identify the optimal derivation individuals for the algorithm, limiting the full utilization of its exploratory power. Furthermore, the derived populations of MMOEA-GPD are also deficient in convergence, which affects its optimization efficiency.

In view of the above problems, this paper innovatively proposes the MMOEA-EGR algorithm, and its main contributions can be summarized as follows:

1 The goal-directed multimodal multi-objective evolutionary algorithm framework has been optimized and upgraded, with an innovative introduction of a reinforcement learning mechanism to replace traditional phase control parameters. In this approach, decision variables are treated as states, different evolutionary stages as actions, and population fitness improvement as the reward, which guides the decision of the optimal evolutionary strategy for the next stage. This method significantly enhances the algorithm's versatility across various test problems, leading to superior evolutionary efficiency and effectiveness when tackling complex issues.

2 An innovative strategy for remodeling the distribution of derived populations is proposed, which further enhances the comprehensive exploration of the decision space by optimizing the distribution and convergence of derived individuals, and significantly improves the algorithm's search efficiency and global search capability in the decision space.

3 An improved local convergence indicator is proposed to identify individuals with exploration potential more effectively by introducing the spatial location information of individual neighborhood solutions, thus guiding the population to deeply explore more potential equivalent solution spaces.

4 A large number of experiments have been conducted on 45 standard test problems and two real application problems, and the results show that the algorithm outperforms seven current mainstream multimodal multi-objective evolutionary algorithms in terms of optimization performance.

The rest of the paper is organized as follows. Section 2 first provides a brief overview of the core idea of MMOEA-GPD and points out its limitations. Subsequently, this section will elaborate the research motivation of this paper, introduce the technical background related to this research, and deeply analyze the rationality and innovation of the proposed algorithm in this paper. Section 3 will provide a comprehensive and in-depth introduction to the MMOEA-EGR proposed in this paper. Section 4, through a series of comparative experiments, we will verify the effectiveness of the MMOEA-EGR algorithm. Section 5 will summarize this research work and propose future research directions.

2. Related work

2.1. Brief introduction to MMOEA-GPD

The MMOEA-GPD algorithm adopts a stage-by-stage progressive evolutionary framework, which is divided into three key phases: convergence phase, population derivation phase and diversity maintenance phase, which cooperate with each other to gradually improve the performance of the algorithm.

Convergence phase: the main goal is to accelerate the convergence process of the population in the early and late stages of evolution, so as to improve the overall convergence efficiency. In this phase, any kind of MOEA with strong convergence ability can be flexibly selected as the optimization tool.

Diversity maintenance phase: aims to guide the population to achieve good distributional characteristics in the decision space and target space. For this purpose, the algorithm adopts a co-evolutionary strategy to construct two complementary populations: one of them evaluates the fitness of individuals based on the crowding distance between the dominance relation and the decision space, and the other evaluates the dominance relation based on the crowding distance between the dominance relation and the goal space, and the two populations co-evolve in order to enhance the diversity.

Population derivation phase: aiming at the two core challenges of multimodal multi-objective optimization, i.e., the difference in search difficulty and uneven allocation of computational resources, the algorithm innovatively designs a strategy in the population derivation stage: by combining Eq. (2) with the congestion distance assessment of individuals, the algorithm filters out individuals with potentials that can guide the population to explore more regions of equivalent PSs. Subsequently, the algorithm generates a certain number of new individuals within the neighborhoods of these potential individuals, thus allocating more computational resources to these subspaces for a deeper and broader search.

$$\mathrm{F}(i) = \mathrm{R}(i) + \mathrm{D}(i) \tag{2}$$

$$\mathrm{R}(i) = \sum_{j \in \mathrm{P} \bigwedge j \succ i} \mathrm{S}(j) \tag{3}$$

$$\mathrm{S}(i) = |\{j \,|\, j \in \mathrm{P} \bigwedge i \succ j\}| \tag{4}$$

$$\mathrm{D}(i) = \frac{1}{\delta_i^k + 2} \tag{5}$$

where the set base operator $|\cdot|$ characterizes the total number of set elements, and individual fitness is jointly determined by the dominance metric $R(i)$ and the distribution metric $D(i)$. Where $R(i)$ quantifies the degree of relative disadvantage of an individual in the population (non-dominated individuals take zero value), and $D(i)$ is used as an auxiliary indicator to solve the problem of convergence of fitness due to non-dominated relationships in the original population, which is calculated by strictly following the method proposed in the literature (Zitzler et al., 2001), i.e., taking $k = \sqrt{N+\bar{N}}$ for density estimation.

As illustrated in Fig. 2, the simplified MMOP comprises four distinct Pareto sets (PSs). Notably, evolutionary search exhibits aggregation tendencies in specific decision subspaces (e.g., PS1 and PS4), where nondominated solutions disproportionately concentrate. While Scheme A shows extensive potential for decision space exploration, it has poor convergence performance and is not easily preserved for the next generation compared to the non-dominated schemes that dominate under the evaluation criteria. On the other hand, individuals represented by Solution B have shown strong convergence ability in guiding the population to explore PS3 in depth, although due to the scarcity of such solutions and their low probability of producing quality solutions as parents, multiple iterations are usually required to fully explore the potential of PS3. The strategy of this paper is precisely to invest more computational resources in these key subspaces to effectively explore potential equivalent PSs.

Although MMOEA-GPD shows good performance, it still has the following limitations:

1 The three-stage switching of the MMOEA-GPD is controlled via fixed parameters and shows insufficient generalizability in the face of diversify MMOPs, as clearly exemplified by the experiments on the IDMP test suite in Section 4.2.3.

2 The population derivation stage helps to reasonably allocate computational resources through the derivation of individuals with exploration potential, more comprehensively explores the potential search space, facilitates efficient search and optimization, and may produce individuals with better quality. However, the convergence of most individuals is poor, and multiple population derivations are required, which affects the overall evolutionary efficiency to a certain extent.

3 The design of the derivation metrics is limited to simply adding the Pareto dominance relationship of individuals with the crowding distance, which is not effective in balancing convergence and diversity. Although more equivalent PSs can be found on most of the MMOPs, this defect limits the exploration ability to be fully utilized, resulting in poor results on some problems (e.g., IDMP test question)

To cope with the above problems, this paper will detail the improvements and related techniques in the following three subsections.

2.2. Shifting evolutionary stages with reinforcement learning

To address the first limitation faced by MMOEA-GPD, existing adaptive techniques or reinforcement learning methods can be leveraged. Credit-based adaptive selection strategies have been widely applied in operator selection (Sallam et al., 2020). Inspired by this, the current study can reward the evolution stage by measuring the overall fitness improvement of the population before and after different evolutionary stages. Based on these rewards, methods such as roulette wheel selection (Zhang & Sanderson, 2009), fuzzy inference systems (Santiago et al., 2018), and dynamic multi-armed bandit algorithm (DaCosta et al., 2008) can be used to select stages.

However, this strategy faces the dilemma of exploration versus exploitation: if the evolutionary stage is directly selected based on recent rewards, the stages with higher rewards will be prioritized, reinforcing positive feedback and causing rewards to continually increase. This could potentially lead to a local optimum. Furthermore, this selection strategy relies on the historical improvement in population fitness to determine the next evolutionary stage, but the stages that performed well in the past may lose their effectiveness or even fail in the future.

Reinforcement learning seeks to determine the optimal solution for dynamic problems with finite or infinite states, guiding agents in decision-making within evolving environments (Sutton & Barto, 1998). In MOEAs, it has been applied to environment selection (Van Moffaert et al., 2013), offspring generation (Li et al., 2020), and change tracking (Zou et al., 2020), yet its use in evolutionary strategy selection remains unexplored.

In reinforcement learning systems, the fundamental goal revolves around formulating an optimal policy π that guides agents to make sequential decisions for long-term reward maximization (Shao et al., 2019). During each timestep t, the agent executes an action a_t following the policy's guidelines in state s_t, subsequently obtaining environmental feedback r_t and progressing to state s_{t+1}. This cycle persists until reaching terminal states, with transition experiences (s_t, a_t, r_t, s_{t+1}) being systematically incorporated into the experience replay buffer. These state-action-reward trajectories serve as critical inputs for policy optimization through iterative neural network updates.

Deep reinforcement learning methods are generally categorized into policy-based and value-based approaches. Policy-based methods directly learn a stochastic policy to address continuous action spaces, using independent function approximators (Sutton et al., 1999). The policy $\pi = p(a_t|s_t;\theta)$ defines the probability of taking action a_t in state s_t given the parameters θ. The agent optimizes the policy through gradient methods and can sample from π (Sun et al., 2021) to select actions.

In contrast, value-based methods approximate the action value function for discrete action spaces, which estimates the expected cumulative reward (Q-value) for taking action a_t in state s_t (Dearden et al., 1998). Specifically, the action value function is $Q(s_t, a_t) = \mathbb{E}[R_t|s_t, a_t]$, where $R_t = \sum_{i=t}^{\infty}\gamma^{i-t}r_i$ represents the discounted sum of future rewards, and $\gamma \in [0,1]$ controls the weight of future rewards. With this approximation, the agent selects the action with the highest Q-value in a given state: $\pi(s) = argmax_{a\in\mathscr{A}}Q(s,a)$.

The classical Q-learning algorithm operates through tabular value iteration, maintaining a multidimensional Q-table (Watkins & Dayan, 1992) that stores reward expectations for each state-action pair. While effective in discrete state representations, this methodology encounters dimensional constraints in continuous spaces. To address this, deep Q-networks (DQN) (Mnih et al., 2015) implement function approximation through deep neural architectures, where environmental states are processed as input tensors to generate corresponding Q-value vectors across the action spectrum. Through agent-environment interactions, transition sequences are accumulated in circular replay memory, facilitating decoupled iterations between exploratory policy execution and batch-based network optimization. Parameter updates are achieved via backpropagation minimization of temporal difference errors using stochastic gradient descent.

$$\mathscr{L} = \frac{1}{\mathscr{T}}\sum_{t\in\mathscr{T}}(Q(s_t,a_t) - q_t)^2 \tag{6}$$

$$q_t = r_t + \gamma_{a'\in\mathscr{A}}^{max}Q(s_{t+1},a') \tag{7}$$

where, $\mathscr{T}$ is the training set, $Q(s_t,a_t)$ denotes the Q −value of the a_t output neuron of the neural network at input vector s_t, and q_t is the Q −value under state r_t. Note that the expected output q_t contains not only the current reward r_t but also the maximum expected reward for the next action, i.e., $max_{a'\in\mathscr{A}}Q(s_{t+1},a')$. Due to this recursive property, q_t can represent the maximum value of the future cumulative reward.

Building on this concept, solution decision variables are treated as states, while candidate stages correspond to actions. Given that states are continuous and actions are discrete, DQN is a more suitable choice for processing. Section 3 will provide a detailed explanation of the proposed method and the implementation of MMOEA.

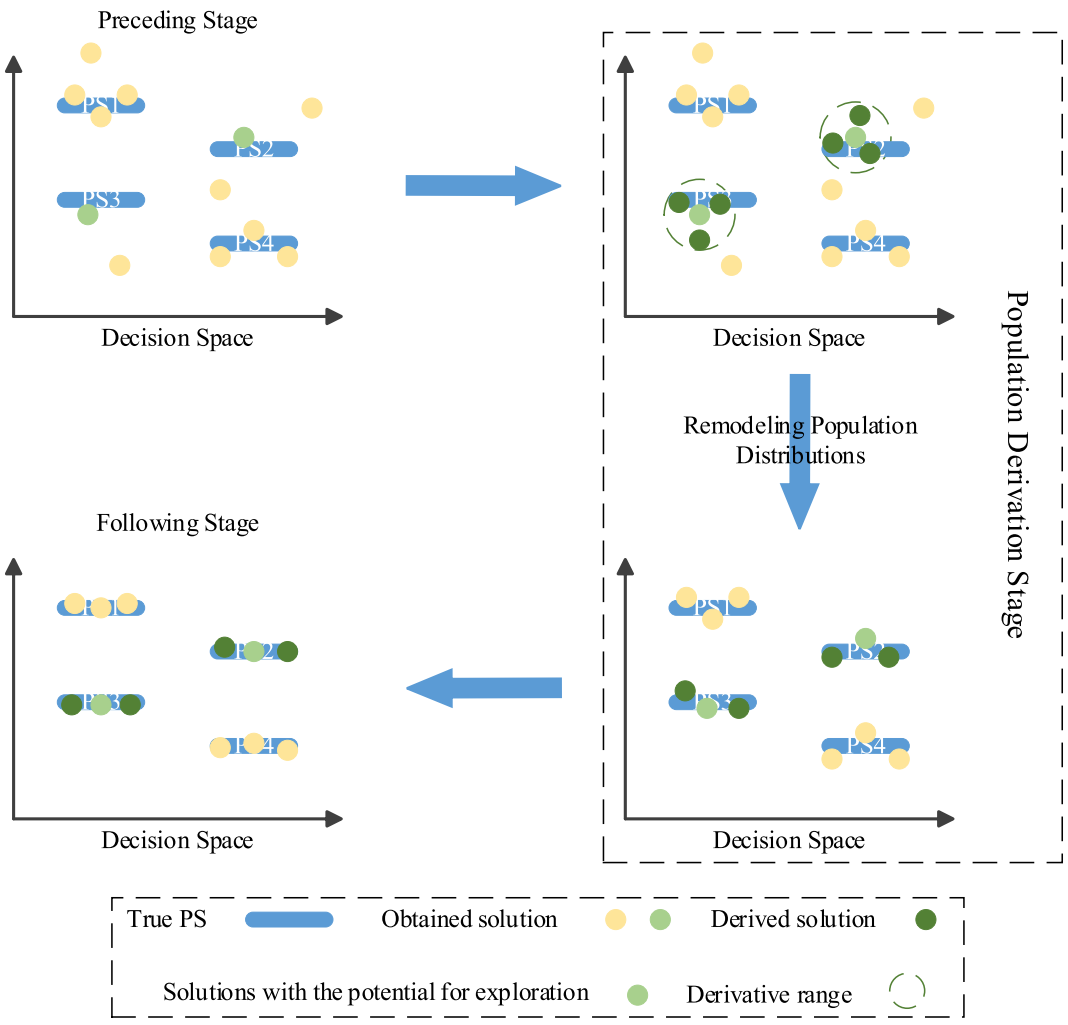


Fig. 3. Schematic diagram of the strategy for remodeling the distribution of populations.

2.3. Remodeling the distribution of derived population

After implementing the population derivation strategy in MMOEA-GPD, the algorithm significantly increases exploration in regions PS2 and PS3, resulting in individuals in these regions having a better chance of being selected as parents, and thus producing higher quality offspring (as shown in Fig. 2(b)). However, it can be observed that although the newly generated derived populations may contain individuals of higher quality, these newly generated individuals are overall less convergence, which affects their competitiveness in environmental selection. After merging with the original population, only a small number of derived individuals enter the next generation, which greatly limits the evolutionary efficiency of the algorithm and the potential utilization of the derived population.

To address this problem, this paper proposes a strategy to remodeling the distribution of derived populations in MMOEA-EGR, aiming to fully exploit the potential value of derived populations. Specifically, the derived population is first regarded as an independent sub-population, and its convergence characteristics are utilized to reselect the parent generation to guide a new round of evolution, so as to form a population distribution that is closer to the true Pareto frontier within the current PS region. The optimized derived population is then merged with the original population to improve the overall evolutionary efficiency of the population (as shown in Fig. 3). This strategy has the following key advantages:

1 Reducing the interference of converging individuals: Treating the derived population as a separate sub-population avoids the interference of individuals in the original population that have already converged in other PS regions, thus enabling the derived population to evolve independently in a specific PS region, and focusing more on improving the quality of the solution in that region.

2 Full utilization of derived individuals: By evolving the derived population independently, the potential of each individual is maximized. Even if the convergence of some individuals is not ideal, their exploration potential in other PS regions can be fully retained and utilized to support the expansion of the algorithm in these areas.

3 Improvement of evolutionary efficiency: This strategy improves the exploration efficiency of the algorithm in different PS regions by effectively balancing convergence and exploration, which enables the improved individuals of the derived population to contribute to the overall population to a greater extent, thus realizing a faster and more comprehensive Pareto solution set approximation.

In summary, remodeling the distribution strategy of the derived populations not only improves the evolutionary efficiency of the algorithm, but also better exploits the exploration potential of the derived populations, which makes the performance of MMOEA-EGR more robust and efficient in the complex problem space.

2.4. Improved local convergence indicator

When addressing the imbalanced distance minimization benchmark problem (IDMP) (Liu et al., 2019), the derived indicator design of

MMOEA-GPD faces a performance degradation issue. This problem arises from the significantly different search difficulties across various equivalent Pareto sets. Solutions near one PS are more easily dominated by solutions near another PS. This happens because simply adding the Pareto dominance and crowding distance of individuals fails to effectively reflect their actual exploration potential.

To address this issue, Li et al. (2021) introduced a weighted IBEA indicator to balance diversity and convergence, ensuring population diversity. Lin et al. (2020a) adopted harmonic mean distance as the crowding indicator in the decision space and combined clustering methods to obtain well-distributed Pareto sets. Li et al. (2022) proposed a local convergence indicator to assess the local convergence of the population, controlling the search and retention of local solutions, thus achieving a balance between diversity and convergence. Building on this, Li et al. (2023) proposed a local convergence indicator that demonstrated excellent performance, specifically:

$$I_{LC}^{i} = \sum_{j \in N_i} S_j \cdot D_{j,i} \tag{8}$$

$$S_i = \sum_{j \in N_i} D_{i,j} \tag{9}$$

where N_i represents the set of neighboring solutions of x_i, and $D_{i,j} = 1$ indicates that x_i dominates x_j. S_i represents the total number of dominated solutions. $I_{LC}^{i} = 0$ indicates that x_i is a non-dominated solution. The range of the neighborhood, R, is determined by the formula (10).

$$R = \frac{\sum_{i,j=1}^{N} d_{i,j}}{2N^2} \tag{10}$$

where $d_{i,j}$ represents the Euclidean distance between solutions x_i and x_j, and R is half of the average distance between all solutions. This range setting provides a quick basic distance threshold suitable for scenarios where the individuals are evenly distributed. However, when individuals are unevenly distributed (e.g., in mid-evolution, populations may be divided into small clusters distributed around different equivalent Pareto sets, or there are outlier individuals), this method may not accurately capture differences in local structure, resulting in some high-quality individuals being overlooked due to outlier individuals or problems with the density of distribution.

To address this, this paper proposes an improved local convergence indicator, I_{KLC}^{i}, which dynamically adjusts the range of R based on the spatial position information of the K nearest neighbors around an individual. The specific calculation formula is given in formula (11).

$$R = \frac{\sum_{i=1}^{N} \sum_{j=1}^{K} d_{i,j}}{N \cdot K} \tag{11}$$

where N is the total number of individuals in the population and K is the number of nearest neighbors of each individual.

The advantages of this division are:

1 Local adaptation: By determining the neighborhood radius of a solution based on the K solutions of the nearest neighbors, the method is better adapted to the situation of uneven distribution within the population.

2. Reducing the influence of outliers: since only K nearest neighbors are considered, the influence of outlier individuals on the average distance is significantly reduced, which makes the calculation of R more robust and less prone to be large or small.

This paper discusses the differences between the method proposed in this study and W. Li's method through a more specific example: in the middle of the evolutionary process, the overall distribution of the population usually shows clustering around multiple different equivalent PSs due to the differences in the searching difficulty of each equivalent PSs. At this time, the method proposed by W. Li will get a larger result when calculating the R value. However, since some of the equivalent PSs are easier to search, a large number of individuals tend to cluster preferentially around them and tend to converge, forming a denser distribution than that around other PSs that are difficult to search. These individuals close to convergence are valuable in their own right, but due to the wide range of R values, they are easily misclassified as inferior individuals in the current evaluation. To address this problem, the algorithm proposed in this paper uses the average distance of K nearest neighbors based on all individuals to calculate the R-value, which is able to better adapt to the uneven distribution of the population and balance the individuals that have approached convergence with those that still have potential for exploration in the evaluation.

3. Proposed methodology

3.1. Reinforcement learning based approach to evolutionary stage selection

When using reinforcement learning, the central issue is how to define the rewards and states of a task in a way that allows an intelligent to learn the relationship between the two using a deep neural network. In order to evaluate the fitness improvement of the population at the end of each evolutionary stage, the proposed method calculates the overall fitness value improvement using Eq. (12) to account for both convergence and diversity.

$$FI\left(FP^{before}, FP^{after}\right) = max\left\{1 - \frac{FP^{after}}{FP^{before}}, 0\right\} \tag{12}$$

$$FP = \sum_{i=1}^{N} I_{KLC}^{i} \tag{13}$$

where FP^{before} and FP^{after} denote the overall fitness value of the population before and after the execution of the stage, respectively, both calculated by Equation (13). In addition, to incorporate historical information, the method records the population fitness improvement FI generated by the stage es by saving the most recent tuple (es, FI) in a first-in-first-out queue R.

$$reword_{es} = \max_{(es,\ FI) \in \mathrm{R}} FI \tag{14}$$

In this approach, the reward for a stage is defined as the highest FI value within the queue. The rationale is that rare yet significant improvements hold greater value than frequent but minor ones (Fialho et al., 2010). Thus, using the maximum FI value instead of the average ensures reward stability across stages over time. Algorithm 1 presents the pseudo-code for this reward allocation strategy.

Algorithm 1: Credit Assignment ($P^{before}, es, P^{after}, R$)

Input: Original Population: P^{before}, Evolutionary Stage: es, Populations after the implementation of evolutionary stage: P^{after}, Queue of rewards: R
Output: Updated Queue: R, Reward of the stage:$reword_{es}$
1: $FI \leftarrow 0$;
2: $[FP^{before}, FP^{after}] \leftarrow$ Calculate the fitness by (13);
3: Add (es, FI) to R;
4: **if** R *exceeds its maximum size*
5: Delete the oldest tuple from R;
6: **end if**
7: $reword_{es} \leftarrow$ Calculate the reward of es by (14);
Return: $reword_{es}$, R;

After generating progeny solutions and computing rewards, the tuple $\left(P_{dec}^{b},\ es,\ reword_{es},\ P_{dec}^{a}\right)$ is stored in another experience playback queue $\mathscr{T}$ for training. The state is represented as the average decision variable of all solutions in the population. (i.e., $s = P_{dec}$) Actions are represented as stages of selection (i.e., $a = es$). Next, the deep neural network $Q(s, a)$ is trained to select stages in the future based on Eqs. (6)-(7) using the experience in the cohort $\mathscr{T}$. Before each stage begins, the neural network computes Q-values based on the input P_{dec} and selects stages using a roulette mechanism. However, if a stage is absent from the

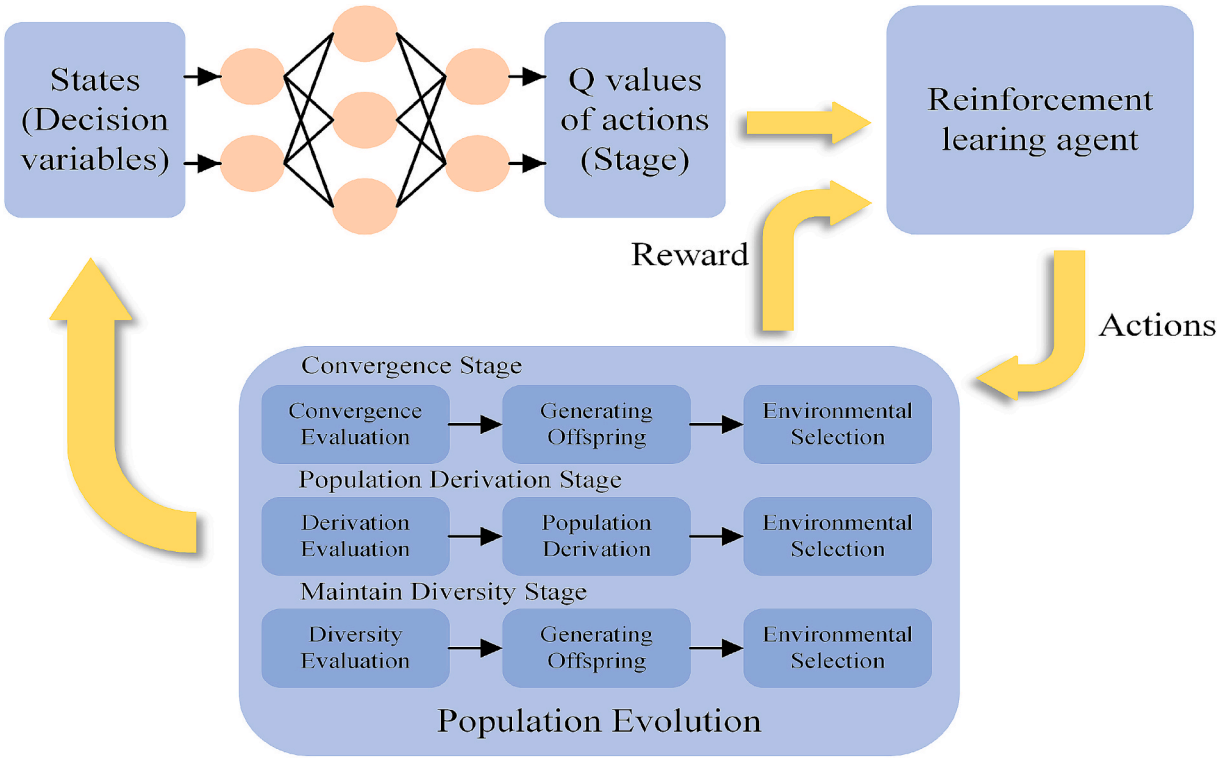


Fig. 4. Illustration of MMOEA-EGR with reinforcement learning based stage selection.

empirical playback pool $\mathscr{T}$, it cannot be chosen through roulette and must be forcibly selected. Algorithm 2 presents the pseudo-code for this stage selection strategy.

Algorithm 2: Evolutionary Stage Selection ($P, ES, \mathscr{T}$, Q)

Input: Population: P, Candidate stage set: ES, Experience replay pool: $\mathscr{T}$, Trained neural network: Q
Output: Selected Stage: es
$P_{dec} \leftarrow$ Calculate the average decision variable for population P;
$state \leftarrow P_{dec}$;
$Qvalue \leftarrow 1 \times |ES|vector\ of\ zeros$;
For es $\in$ ES **do**
if $(\bullet, es, \bullet, \bullet) \notin \mathscr{T}$
return es;
else
$Qvalue_{es} \leftarrow$ Q$(state, es)$;
End
$es \leftarrow$ Select a stage according to $Qvalue$ via roulette-wheel selection;
Return: es;

The stage selection method formulated in this paper adopts the average decision variable of the population as a whole as the state because it can comprehensively capture the distribution and evolutionary trend of the population and provide a global perspective for the intelligences. Moreover, the average decision variable eliminates the noise caused by outliers and provides a stable input for reinforcement learning. This stability is crucial for training deep neural networks, as sharp fluctuations caused by individual solutions can destabilize policy updates. Meanwhile, when calculating rewards, we calculate the overall fitness value improvement through the fitness improvement of each individual, so as to choose the operation stage more rationally. With this design, the algorithm is able to strike a good balance between global trends and individual differences, which makes the process of selecting stages consider the convergence of the population and also retain the diversity. Therefore, it is reasonable to use the average decision variable as the state and the overall fitness value improvement of the population as the reward.

3.2. MMOEA-EGR algorithm

In this paper, MMOEA-EGR is proposed based on the evolutionary stage selection method of reinforcement learning. Algorithm 3 describes the evolutionary process of MMOEA-EGR in detail. First, population initialization is performed and deep neural network is constructed. In each generation of the evolutionary loop, the evolutionary stage performed in the current iteration is first selected according to Algorithm 2, and a new generation of the population is generated based on the evolutionary strategy of the selected stage (see lines 7–8). Subsequently, the reward for that stage is calculated based on Algorithm 1 and deposited into the experience pool $\mathscr{T}$ (see lines 9–10). Finally, the information in the experience pool $\mathscr{T}$ is trained based on Eqs. (6) and (7).

Algorithm 3: MMOEA-EGR

Input: Maximum generations: G_{max}, Population size: N, Candidate stage set: ES
Output: Population P
$P \leftarrow$ Initialization(N);
$g \leftarrow 0$;
Q $\leftarrow$ Randomly initialize a deep neural network;
$R \leftarrow \emptyset$;//*Queue of rewards*
$\mathscr{T} \leftarrow \emptyset$;//*Experience replay pool*
While $g < G_{max}$ **do**
$es \leftarrow$ Evolutionary Stage Selection $(P, ES, \mathscr{T}$, Q);
$P \leftarrow$ Generation of evolved populations based on es;
$[reword_{es}, R] \leftarrow$ Credit Assignment $(P^{before}, es, P^{after}, R)$;
Add $(P^{b}_{dec}, es, reword_{es}, P^{a}_{dec})$ to $\mathscr{T}$;
if $\mathscr{T}$ *exceeds its maximum size*
Delete the oldest tuple from $\mathscr{T}$;
End
Randomly select a batch of tuples from $\mathscr{T}$ and train Q by (6)–(7);
End while
Return: P;

The core idea of MMOEA-EGR is to utilize the framework of reinforcement learning to strengthen the goal orientation, which enables the algorithm to reasonably select suitable evolutionary stages in the current evolutionary process, thus guiding the population to effectively evolve

along the goals of each stage. Unlike MMOEA-GPD, which adopts fixed stage control parameters, MMOEA-EGR is able to cope with diversified MMOPs more flexibly. Fig. 4 illustrates the schematic diagram of the evolutionary process of MMOEA-EGR.

3.3. Evolutionary stages of the MMOEA-EGR

3.3.1. Convergence stage

The primary goal of the convergence stage is to effectively steer the population toward the true Pareto front, achieving strong convergence performance.

The specific implementation details are shown in Algorithm 4. During this stage, any MOEA with superior convergence characteristics can meet the requirements. In this study, we selected the framework of SPEA2 (Zitzler et al., 2001) and used I^i_{KLC} to evaluate the fitness values.

Algorithm 4: Convergence Stage

Input: Population: P, Population size: N
Output: Population P
1: *Fitness* $P \leftarrow$ *CalFitness*(P);// *Based on* I^i_{KLC}
2: *Parent* $P \leftarrow$ *TournamentSelection*(*Fitness* P);
3: *off* $\leftarrow$ *Variation*(*Parent* P);
4: $P \leftarrow$ *EnvironmentalSelection*($P \cup$ *off*)
Return: P;

3.3.2. Stage of population derivation

The specific implementation details of the population derivation stage are shown in Algorithm 5. First, derivation metrics are computed according to formulas. (8), (9), and (11), therefore identifying individuals in the population that can lead the population to explore other regions that may contain equivalent PSs (see lines 1–2 of the algorithm). It is worth noting that there are usually more than one such individual with exploration potential. In this study, the solutions with the same fitness as the previously identified derived guiding individuals are deposited into the set *Next* and the individual with the largest distance is selected as the second derived guiding individual by calculating their Euclidean distance from the derived guiding individual in the decision space to improve the efficiency of the population's full exploration of the decision space (see algorithm lines 3–13).

Subsequently, more new individuals are generated in the neighborhood of the derived guide individual, allocating more computational resources to explore this region in the next iteration, with the expectation of finding the complete PS more efficiently. As described in Section 2.3, the convergence quality of the derived populations is usually low, so they must be reshaped by Algorithm 6 to optimize the distribution of the populations. Finally, it is merged with the original population P after the latter has undergone environmental selection.

Algorithm 5: Population Derivation Stage

Input: Population: P, Population size: N
Output: Population P
1: *Fitness* $P \leftarrow$ *DerivationCalFitness*(P);
2: *Individual* $\leftarrow$ min(*Fitness* P);
3: $P' \leftarrow$ cancel *Individual* in P;
4: *Next* $\leftarrow \varnothing$;
5: **For** $i = 1$ **to** *num*(P')
6: **if** *Fitness*(i) = *Fitness*(*Individual*)
7: *Next* $\leftarrow$ *Next* $\cup$ i;
8: **End**
9: **End**
10: **For** $i = 1$ **to** *num*(*Next*)
11: *Dis* $\leftarrow$ *Caldis*(i);
12: **End**
13: *Individual* $\leftarrow$ *Next*(max(*Dis*)) $\cup$ *Individual*;
14: $P_G \leftarrow$ *PopulationDerivation*(*individual*, N/5);
15: $P_G \leftarrow$ *RemodelingPopulationDistributions*(P_G, N * 2/5);
16: $P \leftarrow$ *EnvironmentalSelection*(P, N *•3/5);
17: $P \leftarrow P \cup P_G$;
Return: P;

It is important to note that the derived population fitness is calculated using Eq. (15), where $\|x_j - x_i\|$ denotes the Euclidean distance between individuals x and y in the decision space, and is normalized. This is because the goal of the derived population is to guide the population to fully explore the subspace within the neighborhood of the derived guiding individual. When remodeling the distribution of derived population, both its convergence and diversity should be taken into account.

$$F^i_{FD} = I^i_{KLC} + \sum_{j \in N} \frac{1}{\|x_j - x_i\|} \tag{15}$$

Algorithm 6: Remodeling Population Distributions

Input: Population: P, Population size: N
Output: Population P
1: *Fitness* $P \leftarrow$ *CalFitness*(P);//*Based on Eq.*(15)
2: *Parent* $P \leftarrow$ *TournamentSelection*(*Fitness* P);
3: *off* $\leftarrow$ *Variation*(*Parent* P);
4: *Union* $\leftarrow$ *off* $\cup$ P;
5: *Fitness Union* $\leftarrow$ *CalFitness*(*Union*);//*Based on Eq.*(15)
6: *IDX* $\leftarrow$ *sort*(*Fitness Union*);
7: $P \leftarrow$ *Union*(*IDX*(1: N));
Return: P;

3.3.3. Diversity maintenance stage

Algorithm 7 illustrates the diversity maintenance stage, which leverages the concept of co-evolution between two populations (Tian et al., 2020; Zeng et al., 2023).

Specifically, the fitness value of the population P is computed by Eq. (15), which aims to guide the population to be more evenly distributed in the decision space. As for the population P′, the calculation of its fitness value $\|x_j - x_i\|$ is used to denote the distance between the solution x and the solution y in the objective space, which aims to promote the uniform distribution of the population in the objective space. This dual-population co-evolution mechanism takes into account the balanced distribution of the decision space and the objective space while maintaining diversity, which helps to improve the comprehensiveness of the optimization process.

The triphasic framework of MMOEA-EGR operates through coordinated interaction of its constituent stages, each governed by specialized optimization targets that collectively drive population evolution toward diversified Pareto-optimal solutions. The algorithm systematically navigates the solution space to exhaustively identify latent equivalent PSs, while maintaining an equilibrium between solution convergence and diversity preservation. This dual optimization mechanism ensures synergistic advantages in both spatial coverage breadth and localized search depth.

Algorithm 7: Diversity Maintenance Stage

Input: Population: P, Population size: N
Output: Population P
1: $P' \leftarrow P$;
2: *Fitness* $P \leftarrow$ *CalFitness*(P);//*Based on Eq.(15)* and Use of decision space congestion distances
3: *Fitness* $P' \leftarrow$ *CalFitness*(*Fitness* P'); //*Based on Eq.(15)* and Use of objective space congestion distances
4: *Parent* $P \leftarrow$ *TournamentSelection*(*Fitness* P);
5: *Parent* $P' \leftarrow$ *TournamentSelection*(*Fitness* P');
6: *off* $P \leftarrow$ *Variation*(*Parent* P);
7: *off* $P' \leftarrow$ *Variation*(*Parent* P');
8: *Union* $\leftarrow P \cup P' \cup$ *off* $P \cup$ *off* P';
9: $P \leftarrow$ *EnvironmentalSelection*(*Union*, N);
Return: P;

Table 1
The average rankings of IGD and IGDX for MMOEA-EGR and each comparison algorithm across the MMF, MMMOP, IDMP, and HYL test suites, with the best results in each row highlighted.

Problem	Indicators	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
MMF	IGD	6.88	5.13	4.25	7.63	5.38	2.50	2.63	**1.63**
	IGDX	4.00	5.13	5.25	8.00	3.75	**2.50**	4.63	2.75
MMMOP	IGD	7.15	5.25	5.25	5.00	4.85	4.85	2.20	**1.45**
	IGDX	6.85	5.60	5.55	3.75	4.80	4.15	2.80	**2.50**
IDMP	IGD	6.58	5.67	5.17	7.92	3.83	3.75	1.75	**1.33**
	IGDX	3.25	4	3.75	7.75	4.25	4	6.83	**2.17**
HYL	IGD	3.80	4.00	6.40	5.40	4.80	5.00	3.60	**2.40**
	IGDX	7.00	5.80	6.00	6.40	3.40	3.80	2.00	**1.60**

4. Experimental results and analysis

4.1. Experimental setup

1) Comparison algorithm.

To assess the performance of MMOEA-EGR in solving multimodal multi-objective optimization problems, this study selects seven mainstream MMOEAs as benchmark algorithms.

- ILC-MMEA (Li et al., n.d.), an algorithm that aims to enhance population diversity by designing local convergence metrics and improving environmental selection strategies for population crowding.
- MMEA-WI (Li et al., 2021) utilizes weighted metrics to preserve promising solutions, thereby improving diversity.
- MMOEA/DC (Lin et al., 2020), an algorithm that employs a dual clustering approach aimed at enhancing the diversity of both the decision space and the objective space.
- Tri-MOEA&TAR (Liu et al., 2018) enhances optimization performance through dual archiving and recombination strategies.
- HREA (Li et al., 2022), which utilizes a hierarchical sorting strategy, aims to further enhance the diversity of the algorithm.
- MMEAPSL (Ming et al., 2024), Learning the topology of PS using GNG neural networks to help the algorithm explore the decision space.
- MMOEA-GPD (Deng et al., 2024), an algorithm that incorporates three evolutionary strategies aimed at improving the search capability of the decision space while maintaining a balance between diversity and convergence.

2) Benchmark problems and metrics.

In this study, four benchmark test suites were selected, including: the widely used MMF benchmark suite (Yue et al., 2017), the MMMOP suite (Liu et al., 2018), which places high demands on the algorithm's search capability and features scalable objective and decision spaces, a new test suite HYL (Ishibuchi et al., 2022a, 2022b) with an infinite Pareto set, and a set of imbalanced distance minimization problems IDMP (Liu et al., 2019), comprising a total of 45 problems.

These problems feature varying numbers of Pareto sets and diverse challenges, enabling a comprehensive assessment of MMOEAs' performance and functionality (Ishibuchi et al., 2022a, 2022b). This study employs two mainstream evaluation metrics, IGD (Zitzler et al., 2003) and IGDX (Zhou et al., 2009), with their calculation formulas provided below.

$$IGD(\mathrm{X}) = \frac{1}{|\mathrm{X}^*|}\sum_{y\in \mathrm{X}^*}\min_{x\in \mathrm{X}}\{ED(f(x),f(y))\} \tag{16}$$

$$IGDX(\mathrm{X}) = \frac{1}{|\mathrm{X}^*|}\sum_{y\in \mathrm{X}^*}\min_{x\in \mathrm{X}}\{ED(x,y)\} \tag{17}$$

where $ED(x,y)$ denotes the Euclidean distance between x and y. x and X represent an obtained individual and the obtained solution set, respectively, while X^* corresponds to the sampled true Pareto nodes. IGD and IGDX measure the closeness of the algorithm's solution set to the true Pareto front and the true Pareto solution set, respectively. Lower values of these metrics indicate a smaller deviation from the true Pareto front or solution set, signifying superior algorithm performance.

3) Parameter settings and genetic operators:

All algorithms adopted Simulated Binary Crossover (SBX) (Deb & Agrawal, 1995) and Polynomial Mutation (PM) (Deb et al., 2002) from Genetic Algorithms (GA) as the variation operators, with the following parameter settings:

- Crossover probability: $p_c = 1$; distribution index: $\eta_c = 20$;
- Mutation probability: $p_m = \frac{1}{n}$; distribution index: $\eta_m = 20$;

In the four test suite experiments, we carefully considered the computational resource requirements of various comparison algorithms based on the original authors' information. We ensured these requirements were met before determining the appropriate population sizes and maximum evaluations. The setup details are as follows:

- MMF: N = 200 E_{max} = 10000 ; .
- MMMOP: N = 400 E_{max} = 50000 ; .
- IDMP: N = 200 E_{max} = 20000 ; .
- HYL: N = 200 E_{max} = 20000 ; .

The parameters of the comparison algorithm were set according to the configurations in the original study. The parameter settings for MMOEA-EGR are as follows: the derivation range and the population size for derivation are consistent with those of MMOEA-GPD. In the improved local convergence indicator I^i_{KLC}, K is set to 3. The structure of the DQN neural network follows the configuration in (Tian et al., 2022), consisting of a 7-layer fully connected network. The size of the input layer is D, the number of decision variables, and the size of the output layer is consistent with the number of candidate stages. The number of neurons in the five hidden layers are 128, 256, 128, 64, and 32, respectively. The network is trained using the Adam optimizer (Kingma & Ba, 2014), with a learning rate of 0.01.

The results for all test problems were averaged over 30 independent runs, with statistical analysis conducted using the Wilcoxon rank-sum test at a significance level of $\alpha = 0.05$. All experiments were executed on the PlatEMO (Tian et al., 2017) platform. To facilitate further research, the code for MMOEA-EGR will be made publicly available.

4.2. Experimental results analysis

Table 1 summarizes the average rankings of MMOEA-EGR and the compared algorithms across four test suites based on IGD and IGDX metrics. In the MMF test suite, MMOEA-EGR excels in IGD, achieving an average ranking of 1.63 and securing the top position, followed by MMEAPSL with a ranking of 2.50. In contrast, the previously proposed ICL-MMEA and Tri-MOEA&TAR exhibit relatively weaker performance on the IGD metric. Regarding IGDX, MMEAPSL ranks first, with MMOEA-EGR closely following in second place.

In the MMOP, IDMP, and HYL test suites, MMOEA-EGR demonstrates a clear overall advantage, achieving the highest rankings in both IGD and IGDX, underscoring its stability and efficiency in solving multi-objective optimization problems.

Notably, due to the complexity of a single Pareto front potentially corresponding to multiple Pareto sets in the objective space, the true Pareto front is transformed in subsequent visual analyses to provide a clearer representation of the PFs generated by different PSs. In the

Table 2
Statistical results of IGD on MMF benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
MMF1	3.6026e−3	3.7339e−3	3.5277e−3	4.6088e−3	3.6060e−3	**2.9796e−3**	3.2630e−3	3.0688e−3
	(1.98e−4) −	(3.35e−4) −	(1.24e−4) −	(9.55e−4) −	(1.34e−4) −	**(9.87e−5) =**	(3.29e−4) −	(2.65e−4)
MMF2	1.7925e−2	1.2102e−2	1.0742e−2	3.3779e−2	1.7488e−2	**8.5419e−3**	1.5884e−2	1.0703e−2
	(9.26e−3) −	(2.51e−3) −	(2.28e−3) =	(4.94e−2) −	(6.06e−3) −	**(1.09e−3) +**	(2.87e−3) −	(9.61e−4)
MMF3	1.4309e−2	1.0731e−2	1.0287e−2	2.1646e−2	1.4953e−2	**7.0455e−3**	1.2545e−2	9.8946e−3
	(3.17e−3) −	(1.41e−3) −	(5.90e−3) −	(1.10e−2) −	(4.26e−3) −	**(1.05e−3) +**	(2.24e−3) −	(7.02e−4)
MMF4	4.7422e−3	3.7770e−3	3.2446e−3	2.8655e−2	3.3130e−3	3.1318e−3	**2.6443e−3**	2.9309e−3
	(2.35e−4) −	(1.94e−4) −	(5.59e−4) −	(6.61e−2) −	(1.28e−4) −	(1.01e−4) −	**(3.05e−4) +**	(3.15e−4)
MMF5	4.1313e−3	3.4905e−3	3.6051e−3	4.6006e−3	3.5342e−3	3.2038e−3	3.1327e−3	**3.1026e−3**
	(2.02e−4) −	(2.20e−4) −	(1.75e−4) −	(2.13e−3) −	(1.86e−4) −	(1.55e−4) −	(2.98e−4) =	**(2.46e−4)**
MMF6	3.9476e−3	3.5076e−3	3.5370e−3	4.4198e−3	3.4959e−3	3.2364e−3	3.0384e−3	**3.0004e−3**
	(2.12e−4) −	(2.94e−4) −	(1.22e−4) −	(2.07e−3) −	(1.86e−4) −	(1.59e−4) −	(3.45e−4) =	**(2.17e−4)**
MMF7	5.1008e−3	3.9808e−3	3.7687e−3	3.9588e−3	3.5008e−3	3.0501e−3	**2.5517e-3**	2.7988e−3
	(3.76e−4) −	(1.94e−4) −	(3.71e−4) −	(1.32e−3) −	(1.69e−4) −	(9.68e−5) −	**(1.20e-4) +**	(2.67e−4)
MMF8	5.1340e−3	3.6310e−3	2.8143e−3	5.0715e−3	3.9391e−3	3.6789e−3	2.3965e−3	**2.1987e−3**
	(2.70e−4) −	(1.94e−4) −	(3.95e−4) −	(6.42e−3) −	(2.56e−4) −	(2.03e−4) −	(5.99e−5) −	**(5.84e−4)**
+/−/=	0/8/0	0/8/0	0/7/1	0/8/0	0/8/0	2/5/1	2/4/2	

Table 3
Statistical results of IGDX on MMF benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
MMF1	3.8488e−2	4.4727e−2	4.5191e−2	7.0863e−2	3.9171e−2	**3.8254e−2**	4.1993e−2	4.7405e−2
	(1.75e−3) +	(2.52e−3) +	(1.67e−3) =	(9.26e−3) −	(1.80e−3) +	**(1.85e−3) +**	(1.71e−3) +	(4.92e−3)
MMF2	3.4839e−2	3.2956e−2	2.0770e−2	7.6021e−2	3.8607e−2	2.1065e−2	3.7792e−2	**1.8748e−2**
	(1.17e−2) −	(7.70e−3) −	(7.62e−3) =	(4.19e−2) −	(9.02e−3) −	(2.90e−3) −	(7.52e−3) −	**(2.73e−3)**
MMF3	3.3226e−2	2.7928e−2	**1.6469e−2**	5.6292e−2	3.3065e−2	2.0050e−2	3.1009e−2	3.0856e−2
	(7.40e−3) −	(5.17e−3) =	**(7.06e−3) +**	(2.11e−2) −	(9.32e−3) =	(5.04e−3) +	(5.83e−3) =	(1.48e−2)
MMF4	2.4004e−2	2.5387e−2	2.8045e−2	8.5391e−2	1.9766e−2	1.9270e−2	2.1773e−2	**1.3501e−2**
	(8.49e−4) −	(1.82e−3) −	(2.66e−3) −	(1.33e−1) −	(8.29e−4) −	(8.67e−4) −	(1.59e−3) −	**(2.48e−3)**
MMF5	6.2919e−2	7.5275e−2	7.6684e−2	1.0894e−1	6.4792e−2	6.4253e−2	7.2200e−2	**5.9233e−2**
	(1.44e−3) −	(3.56e−3) −	(2.62e−3) −	(1.10e−2) −	(2.79e−3) −	(2.46e−3) −	(3.76e−3) −	**(6.56e−3)**
MMF6	5.8673e−2	6.4525e−2	6.7340e−2	8.8698e−2	5.6515e−2	**5.6251e−2**	5.9976e−2	6.5686e−2
	(1.25e−3) +	(2.72e−3) =	(2.14e−3) −	(8.34e−3) −	(1.75e−3) +	**(1.84e−3) +**	(2.08e−3) +	(3.45e−3)
MMF7	2.3470e−2	2.5243e−2	2.8353e−2	4.0432e−2	1.9183e−2	1.9607e−2	1.8322e−2	**1.4450e−2**
	(9.43e−4) −	(2.44e−3) −	(1.67e−3) −	(1.76e−2) −	(1.44e−3) −	(1.11e−3) −	(6.07e−4) −	**(2.53e−3)**
MMF8	5.0116e−2	8.2130e−2	7.1747e−2	3.3596e−1	4.9132e−2	5.3318e−2	8.2971e−2	**3.8953e−2**
	(3.70e−3) −	(1.53e−2) −	(1.82e−2) −	(1.24e−1) −	(5.90e−3) −	(1.10e−2) −	(1.51e−2) −	**(5.00e−3)**
+/−/=	2/6/0	1/5/2	1/5/2	0/8/0	2/5/1	3/5/0	2/5/1	

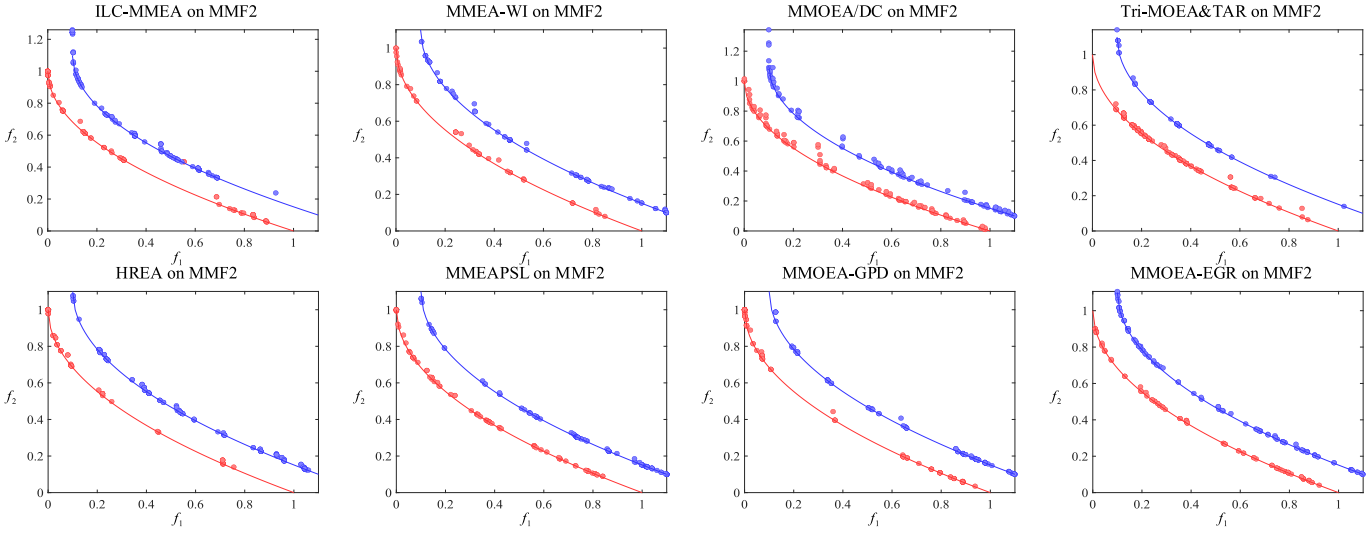


Fig. 5. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the MMF2 problem.

figures, each line corresponds to a PS, ensuring that the number of lines strictly matches the number of PSs. This visualization approach effectively aids in understanding the algorithm's behavior in both decision and objective spaces.

4.2.1. *Analysis of the results of the MMF test suite*

The MMF test suite is a classical benchmark designed to assess the performance of MMOEAs. Table 2 presents the IGD-based experimental results of the proposed algorithm and the comparison algorithms. Both

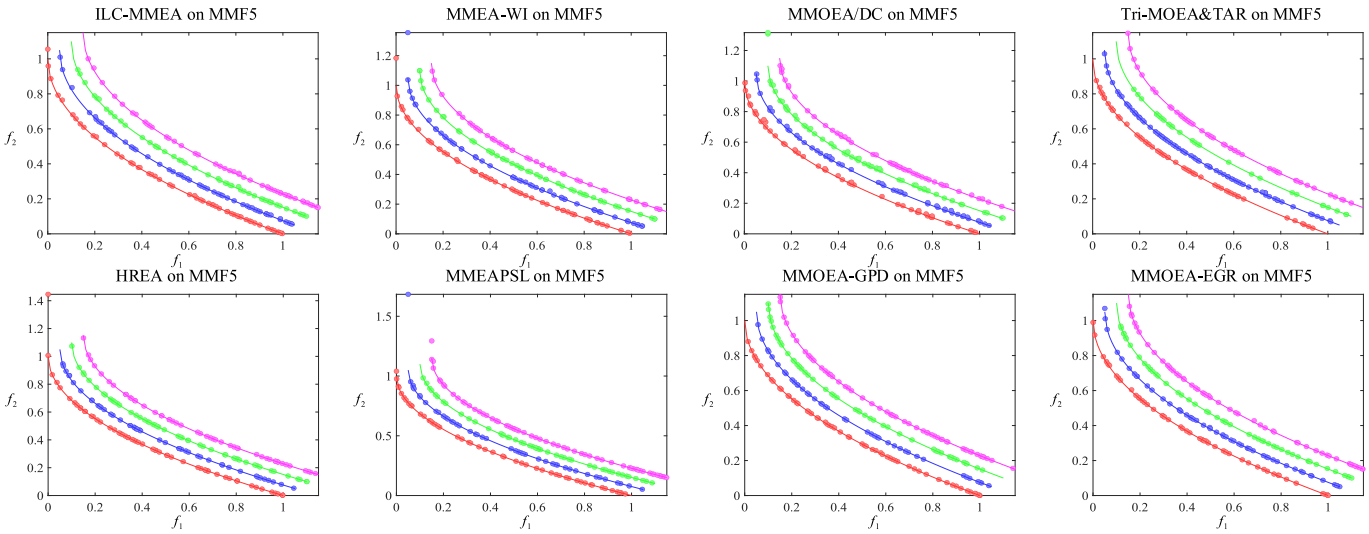


Fig. 6. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the MMF5 problem.

Table 4
Statistical results of IGD on MMMOP benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
MMMOP1_A	1.8546e−3	1.8882e−3	2.1616e−3	2.7340e−3	1.5429e−3	1.4777e−3	1.0452e−3	**1.0155e−3**
	(1.42e−4) −	(2.49e−4) −	(1.85e−3) −	(1.11e−4) −	(1.28e−4) −	(7.17e−5) −	(4.91e−5) −	**(4.93e−5)**
MMMOP1_B	9.5454e + 1	5.4040e−2	1.5484e +	3.2129e−2	2.3577e−1	2.0648e +	4.9922e−2	**2.9661e−2**
	(7.94e + 0) −	(4.40e−2) −	0 (6.98e−1) −	(4.93e−3) −	(3.15e−1) −	0 (7.91e−1) −	(2.01e−2) −	**(7.33e−3)**
MMMOP2_A	2.6953e−3	1.9502e−3	1.3913e−3	3.1595e−3	2.1498e−3	1.8066e−3	**1.0607e−3**	1.4470e−3
	(1.76e−4) −	(1.07e−4) −	(7.20e−5) =	(7.22e−5) −	(1.28e−4) −	(5.77e−5) −	**(1.91e−5)** +	(3.52e−4)
MMMOP2_B	9.2944e−1	5.7424e−2	5.8861e−2	4.5640e−2	5.1968e−2	4.1904e−2	3.3044e−2	**3.1609e−2**
	(4.57e−2) −	(4.91e−3) −	(1.19e−2) −	(6.20e−4) −	(2.34e−3) −	(7.12e−4) −	(2.26e−3) =	**(4.76e−3)**
MMMOP3_A	2.7646e−3	1.9835e−3	1.2278e−3	3.5087e−3	1.9285e−3	1.8306e−3	**1.2041e−3**	1.6667e−3
	(3.17e−4) −	(1.15e−4) −	(1.23e−5) +	(4.07e−5) −	(1.01e−4) −	(6.78e−5) =	**(2.55e−5)** +	(3.43e−4)
MMMOP3_B	1.1774e +	3.9135e−2	7.1196e−2	4.1524e−2	4.6251e−2	3.8238e−2	3.3809e−2	**3.2500e−2**
	0 (1.13e−2) −	(1.09e−3) −	(1.84e−3) −	(4.79e−4) −	(1.79e−3) −	(7.58e−4) −	(7.22e−4) −	**(3.17e−3)**
MMMOP3_C	6.7407e−1	2.5587e−3	3.3005e−3	2.3712e−3	2.8341e−3	1.9155e−3	**1.6760e−3**	2.0920e−3
	(3.63e−2) −	(2.18e−4) −	(2.53e−4) −	(7.95e−5) −	(2.58e−4) −	(8.28e−5) =	**(1.84e−4)** +	(3.66e−4)
MMMOP3_D	1.2122e +	3.9452e−2	6.9264e−2	4.0148e−2	4.5833e−2	3.6598e−2	3.4146e−2	**3.2591e−2**
	0 (1.43e−2) −	(1.24e−3) −	(5.12e−3) −	(5.14e−4) −	(1.69e−3) −	(7.38e−4) −	(1.60e−3) −	**(1.21e−3)**
MMMOP4_A	2.1641e−3	2.3019e−3	1.4176e−3	3.2901e−3	1.5960e−3	1.6398e−3	**1.1964e−3**	1.3740e−3
	(3.41e−4) −	(1.53e−4) −	(3.93e−4) =	(1.09e−4) −	(4.15e−5) −	(3.62e−5) −	**(1.46e−4)** +	(1.57e−4)
MMMOP4_B	1.0035e + 2	3.9300e−1	1.3515e + 1	1.5064e−1	6.1328e−1	1.0865e + 1	1.4428e−1	**1.0850e−1**
	(2.96e−1) −	(3.37e−1) −	(4.07e + 0) −	(4.92e−2) −	(6.45e−1) −	(2.88e + 0) −	(8.18e−2) −	**(8.84e−2)**
MMMOP4_C	7.0958e + 1	1.7309e−1	2.4672e−1	2.3360e−2	1.6318e−1	3.3977e +	2.4828e−2	**4.4667e−3**
	(5.70e + 0) −	(1.36e−1) −	(1.89e−1) −	(2.03e−2) −	(3.50e−1) −	0 (1.03e + 0) −	(1.46e−2) −	**(2.12e−3)**
MMMOP4_D	9.8700e + 1	2.1366e−1	5.2803e +	9.5406e−2	4.5940e−1	5.6725e +	6.3443e−1	**4.8997e−2**
	(2.70e + 0) −	(1.33e−1) −	0 (2.36e + 0) −	(3.66e−2) −	(5.57e−1) −	0 (2.09e + 0) −	(5.72e−1) −	**(1.77e−2)**
MMMOP5_A	1.9438e−3	1.8205e−3	1.4579e−3	3.0915e−3	1.3810e−3	1.5755e−3	**1.0633e−3**	1.2105e−3
	(3.31e−4) −	(5.95e−5) −	(4.51e−4) −	(1.02e−4) −	(2.26e−5) −	(5.93e−5) −	**(8.13e−6)** +	(1.22e−4)
MMMOP5_B	9.9662e + 1	1.3262e−1	2.4802e +	5.0894e−2	2.8537e−1	6.4417e +	7.9732e−2	**3.9203e−2**
	(2.93e + 0) −	(1.08e−1) −	0 (1.15e + 0) −	(1.28e−2) −	(4.33e−1) −	0 (2.07e + 0) −	(8.57e−2) −	**(3.05e−2)**
MMMOP5_C	6.5925e + 1	1.3136e−1	5.0846e−2	1.1101e−2	1.2453e−1	2.7480e +	1.9202e−2	**2.5462e−3**
	(7.01e + 0) −	(1.03e−1) −	(3.25e−2) −	(5.59e−3) −	(3.68e−1) −	0 (1.12e + 0) −	(2.15e−2) −	**(9.38e−4)**
MMMOP5_D	9.8040e + 1	9.2735e−2	6.2117e−1	4.3532e−2	2.3652e−1	3.2549e +	3.6438e−2	**3.4138e−2**
	(5.90e + 0) −	(4.30e−2) −	(4.13e−1) −	(5.84e−3) −	(3.48e−1) −	0 (1.37e + 0) −	(6.11e−3) =	**(3.13e−3)**
MMMOP6_A	1.8792e−3	1.7824e−3	2.0451e−3	1.4552e−2	1.5219e−3	1.4162e−3	**1.1533e−3**	1.3426e−3
	(3.90e−5) −	(5.72e−5) −	(6.95e−5) −	(4.43e−2) −	(4.27e−5) −	(2.49e−5) −	**(1.79e−5)** +	(6.87e−5)
MMMOP6_B	3.9711e−2	3.5887e−2	3.0280e−2	4.4907e−2	3.1619e−2	2.9945e−2	2.6988e−2	**2.5029e−2**
	(7.87e−4) −	(8.72e−4) −	(5.38e−4) −	(5.15e−4) −	(3.95e−4) −	(3.34e−4) −	(1.50e−4) −	**(2.35e−3)**
MMMOP6_C	4.5091e−3	1.8571e−2	4.6664e−3	1.7858e−2	5.4261e−3	9.7956e−3	4.3478e−3	**3.5444e−3**
	(7.35e−4) −	(8.55e−3) −	(7.31e−4) −	(4.24e−2) −	(1.17e−3) −	(2.27e−3) −	(1.11e−3) −	**(1.25e−3)**
MMMOP6_D	6.1601e−2	8.4888e−2	3.9445e−2	4.8273e−2	5.6228e−2	5.2310e−2	3.5799e−2	**3.3982e−2**
	(3.90e−3) −	(1.75e−2) −	(2.69e−3) −	(3.62e−3) −	(1.02e−2) −	(5.60e−3) −	(1.86e−3) −	**(2.61e−3)**
+/−/=	0/20/0	0/20/0	1/17/2	0/20/0	0/20/0	0/18/2	6/12/2	

MMEAPSL and MMOEA-EGR achieve the best results on three problems, while MMOEA-GPD performs exceptionally well on two problems.

In the evaluation based on the IGDX metric (see Table 3), MMOEA-EGR shows a significant advantage, achieving the best results on all five test problems, further highlighting its comprehensive performance in multi-objective optimization problems.

Figs. 5 and 6 display the final Pareto fronts obtained by each algorithm on the MMF2 and MMF5 test problems, respectively. On the

Table 5
Statistical results of IGDX on MMMOP benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
MMMOP1_A	3.7744e−2	4.8340e−2	7.7449e−3	**1.9335e−3**	2.2202e−2	6.0912e−3	9.7581e−3	2.0935e−3
	(3.15e−2) −	(3.29e−2) −	(3.38e−3) −	**(7.82e−5)** =	(1.94e−2) −	(6.25e−3) −	(1.34e−2) −	(8.99e−4)
MMMOP1_B	9.8215e−1	1.8482e−1	2.5874e−1	9.6097e−2	3.0368e−1	2.3267e−1	1.7202e−1	**7.8429e−2**
	(1.12e−2) −	(2.61e−2) −	(2.80e−2) −	(6.78e−2) =	(1.00e−1) −	(2.46e−2) −	(2.66e−2) −	**(1.76e−2)**
MMMOP2_A	3.4643e−3	3.8103e−3	2.9598e−3	2.2432e−2	2.5229e−3	2.3686e−3	**2.3393e−3**	3.2093e−3
	(8.65e−4) −	(1.70e−3) −	(3.47e−4) +	(2.98e−3) −	(1.42e−3) =	(9.10e−4) =	**(4.72e−4)** =	(2.18e−3)
MMMOP2_B	8.7710e−1	4.5801e−2	1.0517e−1	**3.0602e−2**	5.1332e−2	3.9148e−2	3.7269e−2	4.7857e−2
	(6.14e−3) −	(1.26e−2) =	(2.49e−2) −	**(4.70e−3)** +	(1.16e−2) =	(4.46e−3) +	(3.95e−3) +	(8.01e−3)
MMMOP3_A	8.4032e−3	5.2904e−3	6.3280e−3	**8.7519e−4**	4.5332e−3	5.0573e−3	4.0713e−3	4.1866e−3
	(8.89e−4) −	(2.67e−4) −	(2.65e−4) −	**(5.44e−6)** +	(1.80e−4) −	(2.29e−4) −	(1.65e−4) +	(2.39e−4)
MMMOP3_B	1.0679e +	7.7977e−2	2.1440e−1	2.7438e−2	1.1900e−1	8.4966e−2	9.5257e−2	**2.4732e−2**
	0 (5.73e−3) −	(2.28e−3) −	(8.85e−3) −	(2.23e−3) −	(2.68e−3) −	(3.13e−3) −	(2.80e−3) −	**(5.03e−3)**
MMMOP3_C	8.0010e−1	1.0780e−1	1.0395e−1	**6.9268e−3**	4.7860e−2	7.0361e−3	1.8115e−2	7.0259e−3
	(1.89e−2) −	(3.64e−2) −	(3.79e−2) −	**(1.53e−3)** =	(3.75e−2) −	(4.80e−4) =	(1.25e−3) −	(1.06e−3)
MMMOP3_D	9.7666e−1	8.1992e−2	2.2360e−1	2.9996e−2	1.1843e−1	7.5363e−2	9.9067e−2	**2.5557e−2**
	(4.23e−3) −	(1.78e−3) −	(1.38e−2) −	(2.99e−3) −	(3.83e−3) −	(1.91e−3) −	(3.36e−3) −	**(3.33e−3)**
MMMOP4_A	8.5617e−4	1.0461e−3	2.3753e−3	1.0440e−3	**6.5474e−4**	6.6071e−4	6.8926e−4	6.7140e−4
	(1.38e−4) −	(4.50e−5) −	(8.14e−4) −	(1.27e−5) −	**(6.92e−6)** +	(5.15e−6) +	(1.86e−5) −	(1.48e−5)
MMMOP4_B	1.0670e +	1.7360e−1	3.5379e−1	1.4608e−1	1.6785e−1	2.8651e−1	1.0091e−1	**8.2098e−2**
	0 (9.07e−3) −	(4.98e−2) −	(5.11e−2) −	(2.71e−2) −	(6.16e−2) −	(3.01e−2) −	(2.16e−2) −	**(5.13e−2)**
MMMOP4_C	7.8115e−1	1.8090e−1	2.8196e−2	1.2349e−1	2.8442e−1	1.4227e−1	**1.5792e−2**	6.6445e−2
	(1.45e−2) −	(6.55e−2) −	(1.49e−2) =	(1.14e−1) −	(3.46e−2) −	(2.70e−2) −	**(1.67e−2)** +	(8.44e−2)
MMMOP4_D	9.7252e−1	1.8310e−1	2.6227e−1	2.0638e−1	2.1873e−1	2.5505e−1	1.4008e−1	**9.4121e−2**
	(6.03e−3) −	(5.87e−2) −	(3.64e−2) −	(1.11e−1) −	(8.33e−2) −	(3.49e−2) −	(3.59e−2) −	**(5.05e−2)**
MMMOP5_A	9.3343e−4	1.0693e−3	1.2154e−3	1.0608e−3	**6.5304e−4**	8.8138e−4	6.8665e−4	6.7323e−4
	(1.62e−4) −	(4.86e−5) −	(7.20e−5) −	(2.16e−5) −	**(6.26e−6)** +	(1.51e−5) −	(6.47e−6) −	(1.21e−5)
MMMOP5_B	1.0657e +	1.5893e−1	2.1709e−1	9.6744e−2	1.1334e−1	2.5165e−1	9.5827e−2	**2.7018e−2**
	0 (1.58e−2) −	(5.78e−2) −	(3.55e−2) −	(1.89e−2) −	(4.89e−2) −	(2.81e−2) −	(2.73e−2) −	**(1.56e−2)**
MMMOP5_C	7.6116e−1	2.4238e−1	1.5225e−2	1.2593e−1	2.8518e−1	1.3217e−1	**8.8943e-3**	5.8391e−2
	(2.18e−2) −	(8.43e−2) −	(3.20e−2) =	(1.27e−1) −	(3.49e−2) −	(3.78e−2) −	**(2.06e-3)** +	(8.84e−2)
MMMOP5_D	9.6990e−1	1.8068e−1	1.7027e−1	1.4190e−1	2.4585e−1	2.1015e−1	9.2608e−2	**4.0313e−2**
	(1.27e−2) −	(5.08e−2) −	(2.66e−2) −	(1.07e−1) −	(1.03e−1) −	(3.24e−2) −	(1.76e−2) −	**(1.46e−2)**
MMMOP6_A	4.0758e−3	4.3083e−3	5.3380e−3	1.5313e−2	3.3161e−3	3.1435e−3	**3.1007e−3**	3.4251e−3
	(9.71e−5) −	(1.47e−4) −	(1.88e−4) −	(3.91e−2) −	(8.29e−5) +	(1.13e−4) +	**(5.86e−5)** +	(2.25e−4)
MMMOP6_B	5.5590e−2	4.7175e−2	5.9843e−2	5.4007e−2	**3.7808e−2**	3.8730e−2	4.1379e−2	4.1897e−2
	(1.00e−3) −	(1.28e−3) −	(2.47e−3) −	(1.44e−3) −	**(3.20e−4)** +	(2.85e−4) +	(6.19e−4) =	(1.16e−3)
MMMOP6_C	4.2153e−2	2.3943e−1	2.5138e−2	1.3717e−1	1.0918e−1	8.2630e−2	6.8012e−2	**2.2421e−2**
	(2.36e−2) −	(5.12e−2) −	(4.27e−3) −	(8.02e−2) −	(3.85e−2) −	(1.69e−2) −	(1.58e−2) −	**(7.34e−3)**
MMMOP6_D	1.3424e−1	2.3548e−1	1.2928e−1	1.4839e−1	3.6154e−1	1.5346e−1	**1.1579e−1**	1.7182e−1
	(6.51e−3) =	(3.64e−2) −	(3.89e−3) =	(8.28e−2) =	(9.04e−2) −	(3.05e−2) =	**(3.60e−3)** +	(7.10e−2)
+/−/=	0/19/1	0/19/1	1/16/3	2/14/4	4/14/2	4/13/3	6/12/2	

MMF2 problem, MMEAPSL demonstrates the best convergence performance. Although MMOEA-EGR's convergence is slightly inferior to that of MMEAPSL, it performs well in terms of diversity, especially showing a better balance in the distribution of the two equivalent Pareto sets. On the MMF5 problem, MMEAPSL achieves the best performance in both diversity and convergence. Compared to MMOEA-GPD, the improved algorithm shows significant improvement in the distribution of different PSs. This is made possible by the remodeling population distribution strategy devised in this paper, which makes the derived population fit the distribution of the real PF more efficiently, thus alleviating the

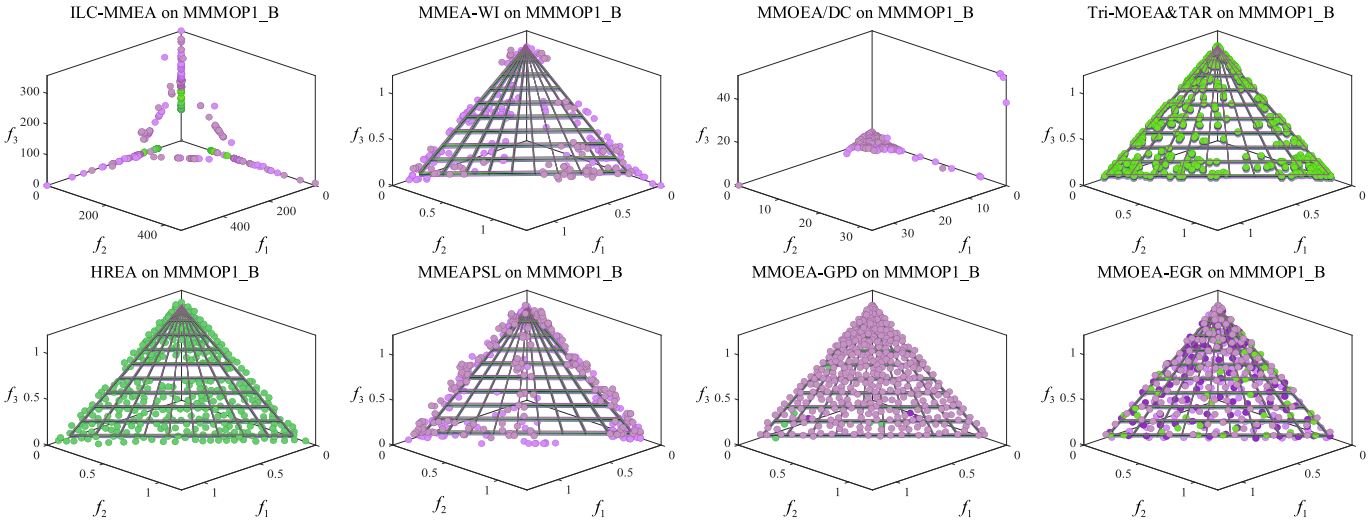


Fig. 7. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the MMMOP1_B problem.

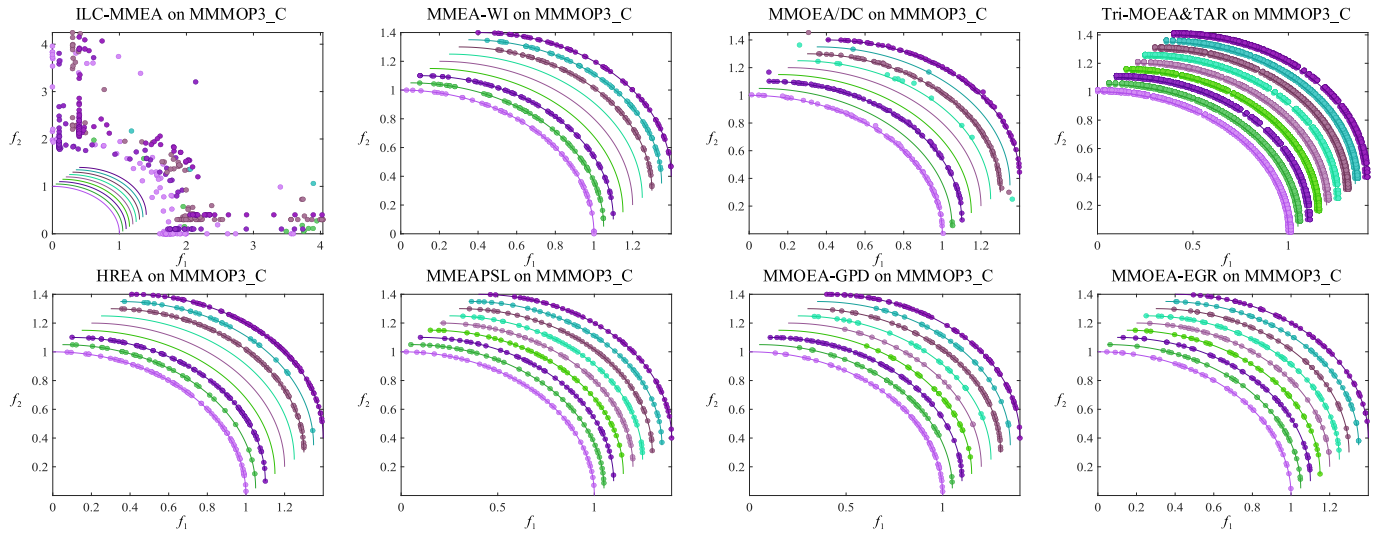


Fig. 8. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the MMMOP3_C problem.

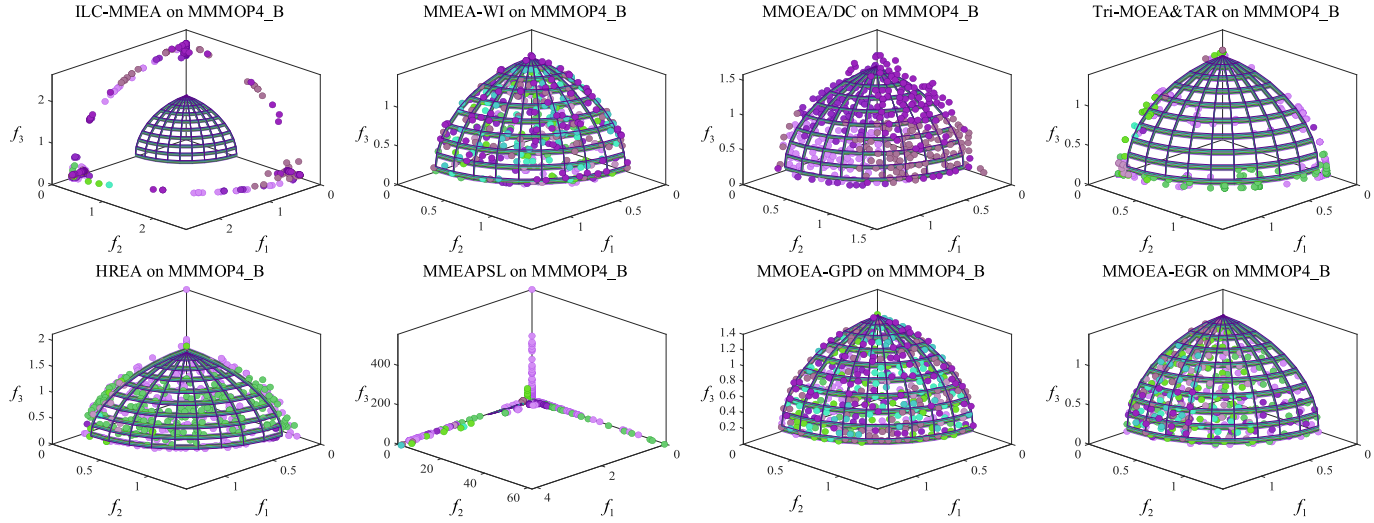


Fig. 9. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the MMMOP4_B problem.

problem of population crowding in localized regions of the search space.

4.2.2. *Analysis of the results of the MMMOP test suite*

The MMMOP benchmark suite integrates multimodal multi-objective problems characterized by variable dimensionality (2–7 decision variables) and dual-to-triple objective configurations. Key test instances contain multiple isomorphic Pareto solution sets, thereby rigorously testing the decision-space exploration capacity of multimodal multi-objective evolutionary algorithms (MMEAs) through intensified landscape complexity.

For this test suite, Table 4 presents the performance of each algorithm based on the IGD metric. The results indicate that MMOEA-EGR demonstrates exceptional performance, achieving the best results in 14 problems, while MMOEA-GPD secures the top results in 6 problems and performs relatively well. Similarly, Table 5 reports the evaluation results based on the IGDX metric, where MMOEA-EGR attains the best results in 8 problems, with MMOEA-GPD also exhibiting strong performance.

Figs. 7–9 show the final Pareto frontiers obtained by the populations of each algorithm in three representative problems, namely MMMOP1_B, MMMOP3_C and MMMOP4_B, respectively. In the MMMOP1_B problem, Tri-MOEA&TAR shows good convergence, but falls short in the number of equivalent PSs found compared to the MMOEA-EGR proposed in this paper. Although MMOEA-GPD finds more equivalent PSs and maintains a certain degree of diversity, the quality of its population distribution still falls short of that of MMOEA-EGR.

In the MMMOP3_C problem, MMOEAPSL learns the topology of PSs through GNG neural networks, while Tri-MOEA&TAR excels in diversity with the help of dual archiving and reorganization strategies. In contrast, MMOEA-EGR uses deep neural networks to optimize the selection of evolutionary strategies and reshape the distribution of derived populations to achieve a significant improvement in population diversity, which is more advantageous than MMOEA-GPD.

In the MMMOP4_B problem, MMOEA-EGR successfully balances diversity and convergence, and achieves significant improvement in overall performance compared to MMOEA-GPD, further verifying the effectiveness of its algorithm design.

4.2.3. *Analysis of the results of the IDMP test suite*

The IDMP test suite consists of scalable and unbalanced multimodal multi-objective optimization problems, characterized by varying difficulty in locating different sets of equivalent Pareto solutions.

Table 6 presents the IGD-based experimental results of the proposed algorithm and the comparison algorithms. MMOEA-EGR achieves the

Table 6
Statistical results of IGD on IDMP benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
IDMPM2T1	5.6523e−4	6.5423e−4	5.0979e−4	6.6762e−4	4.9866e−4	4.9037e−4	**3.9875e−4**	4.0129e−4
	(7.57e−5) −	(2.81e−5) −	(8.20e−5) −	(1.48e−5) −	(5.37e−5) −	(5.42e−5) −	**(4.35e−5)** =	(2.13e−5)
IDMPM2T2	6.1889e−4	6.5283e−4	4.7280e−4	6.6516e−4	5.2286e−4	4.8228e−4	4.7084e−4	**4.4988e−4**
	(3.13e−5) −	(3.27e−5) −	(4.31e−5) −	(3.11e−5) −	(2.20e−5) −	(1.00e−5) −	(2.43e−5) −	**(2.97e−5)**
IDMPM2T3	6.8933e−4	6.7277e−4	5.7619e−4	2.0087e−3	5.5843e−4	4.8416e−4	4.8119e−4	**4.4955e−4**
	(3.16e−5) −	(3.04e−5) −	(9.92e−5) −	(7.36e−3) −	(1.88e−5) −	(1.45e−5) −	(2.57e−5) −	**(3.87e−5)**
IDMPM2T4	5.2921e−4	6.7279e−4	5.4841e−4	6.6785e−4	4.7428e−4	4.6445e−4	**3.8271e−4**	3.9974e−4
	(5.40e−5) −	(2.96e−5) −	(1.50e−4) −	(3.59e−5) −	(4.13e−5) −	(4.49e−5) −	**(2.73e−5)** =	(3.31e−5)
IDMPM3T1	8.3831e−3	6.1907e−3	6.3836e−3	1.1269e−2	5.7655e−3	5.6097e−3	**5.0682e−3**	5.1533e−3
	(3.17e−4) −	(1.44e−4) −	(3.11e−4) −	(7.64e−4) −	(1.55e−4) −	(1.29e−4) −	**(2.57e−4)** =	(2.37e−4)
IDMPM3T2	8.3870e−3	6.2615e−3	5.9037e−3	1.1086e−2	5.6582e−3	5.3796e−3	5.0918e−3	**5.0915e−3**
	(3.10e−4) −	(2.00e−4) −	(1.66e−4) −	(8.26e−4) −	(1.27e−4) −	(1.20e−4) −	(2.67e−4) =	**(1.99e−4)**
IDMPM3T3	9.5964e−3	6.2562e−3	6.4665e−3	1.0882e−2	5.6782e−3	5.3452e−3	5.3646e−3	**4.9858e−3**
	(4.29e−4) −	(1.74e−4) −	(1.80e−4) −	(9.78e−4) −	(1.38e−4) −	(1.51e−4) −	(1.93e−4) −	**(1.80e−4)**
IDMPM3T4	8.1468e−3	6.1939e−3	6.0464e−3	1.1549e−2	5.6047e−3	5.3773e−3	4.8562e−3	**4.8368e−3**
	(3.09e−4) −	(2.30e−4) −	(3.91e−4) −	(7.98e−4) −	(1.36e−4) −	(1.71e−4) −	(2.44e−4) =	**(2.19e−4)**
IDMPM4T1	1.3318e−2	9.7521e−3	1.1583e−2	2.7407e−2	9.4420e−3	8.4031e−3	**6.8022e−3**	7.1039e−3
	(9.43e−4) −	(4.43e−4) −	(1.44e−3) −	(1.68e−3) −	(1.27e−3) −	(1.23e−3) −	**(2.50e−4)** =	(6.40e−4)
IDMPM4T2	1.2347e−2	9.0576e−3	9.0567e−3	2.7193e−2	8.0509e−3	1.0532e−2	7.0117e−3	**6.7600e−3**
	(7.99e−4) −	(6.29e−4) −	(8.20e−4) −	(1.65e−3) −	(1.45e−3) −	(2.64e−3) −	(3.83e−4) −	**(3.01e−4)**
IDMPM4T3	1.2879e−2	8.9738e−3	1.0660e−2	3.5708e−2	7.9549e−3	1.0618e−2	7.3900e−3	**7.2765e−3**
	(6.29e−4) −	(3.15e−4) −	(1.15e−3) −	(2.02e−2) −	(9.88e−4) −	(2.41e−3) −	(5.80e−4) =	**(4.28e−4)**
IDMPM4T4	1.1905e−2	9.0664e−3	9.5228e−3	2.7587e−2	8.3753e−3	1.5690e−2	6.8359e−3	**6.7812e−3**
	(8.52e−4) −	(6.03e−4) −	(1.04e−3) −	(1.86e−3) −	(2.81e−3) −	(4.17e−3) −	(3.50e−4) =	**(3.24e−4)**
+/−/=	0/12/0	0/12/0	0/12/0	0/12/0	0/12/0	0/12/0	0/4/8	

Table 7
Statistical results of IGDX on IDMP benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
IDMPM2T1	**5.5666e−4**	8.2713e−4	6.2119e−4	6.0764e−1	8.3131e−4	6.9504e−4	4.9698e−1	9.8462e−4
	(4.33e−5) +	(3.87e−5) +	(2.57e−5) =	(2.00e−1) −	(1.23e−4) =	(2.05e−4) =	(2.97e−1) −	(1.59e−3)
IDMPM2T2	1.0960e−3	1.0178e−3	1.0737e−4	5.1842e−1	8.1652e−4	6.7677e−4	4.0716e−1	**6.6132e−4**
	(7.97e−5) −	(6.28e−5) −	(5.88e−5) −	(2.86e−1) −	(5.39e−5) −	(3.97e−5) =	(3.32e−1) −	**(4.36e−5)**
IDMPM2T3	1.4418e−3	1.4717e−3	1.3348e−3	1.5199e−1	**1.2928e−3**	1.3665e−3	1.4382e−1	1.6363e−3
	(1.02e−4) +	(1.03e−4) +	(2.14e−4) =	(2.66e−1) −	**(1.26e−4)** =	(4.00e−4) =	(2.70e−1) −	(2.22e−3)
IDMPM2T4	5.4806e−4	4.5740e−2	1.3513e−1	6.0602e−1	6.8462e−2	1.1286e−1	5.8775e−1	**5.4226e−4**
	(1.76e−5) =	(1.71e−1) −	(2.74e−1) −	(2.05e−1) −	(2.05e−1) −	(2.55e−1) −	(2.22e−1) −	**(2.91e−5)**
IDMPM3T1	1.1506e−2	9.3639e−3	9.8088e−3	7.1612e−1	8.1383e−3	**5.4870e−3**	5.1194e−1	8.0502e−3
	(1.91e−4) −	(2.11e−4) −	(3.17e−4) −	(2.58e−1) −	(1.34e−4) −	**(6.27e−4)** +	(2.39e−1) −	(4.59e−4)
IDMPM3T2	1.1640e−2	9.6067e−3	1.7903e−2	5.9237e−1	8.2265e−3	4.2396e−2	4.3011e−1	**8.0217e−3**
	(2.85e−4) −	(2.25e−4) −	(4.47e−2) −	(3.00e−1) −	(1.67e−4) −	(8.50e−2) −	(2.72e−1) −	**(2.49e−4)**
IDMPM3T3	1.2829e−2	1.0104e−2	3.0110e−2	5.2408e−1	8.8689e−3	4.3479e−2	2.7076e−1	**8.1795e−3**
	(3.57e−4) −	(2.88e−4) −	(6.29e−2) −	(2.80e−1) −	(2.18e−4) −	(8.51e−2) −	(2.24e−1) −	**(5.55e−4)**
IDMPM3T4	1.4566e−2	1.1476e−1	8.0184e−2	8.0047e−1	8.1025e−3	1.3228e−1	6.5766e−1	**8.0369e−3**
	(1.76e−2) −	(1.22e−1) −	(1.03e−1) −	(2.81e−1) −	(2.01e−4) =	(1.25e−1) −	(2.94e−1) −	**(3.14e−4)**
IDMPM4T1	1.2579e−2	3.7219e−2	1.7372e−1	1.1424e +	9.2017e−1	6.9039e−1	1.1391e +	**1.1908e−2**
	(3.53e−4) −	(8.00e−2) −	(1.61e−1) −	0 (1.54e−1) −	(3.17e−1) −	(2.88e−1) −	0 (1.45e−1) −	**(1.20e−3)**
IDMPM4T2	**2.9981e−2**	3.5662e−1	2.9177e−1	1.0946e +	1.0136e +	3.7245e−1	9.8115e−1	4.3583e−2
	(4.50e−2) +	(3.25e−1) −	(1.98e−1) −	0 (1.67e−1) −	0 (2.79e−1) −	(2.62e−1) −	(2.13e−1) −	(6.50e−2)
IDMPM4T3	2.7432e−2	1.1124e−1	1.0411e−1	8.9692e−1	9.5004e−1	2.5011e−1	9.0244e−1	**2.6534e−2**
	(2.76e−2) −	(2.39e−2) =	(2.22e−2) =	(3.12e−1) −	(3.48e−1) −	(2.51e−1) −	(2.57e−1) −	**(4.27e−3)**
IDMPM4T4	5.4107e−2	4.4045e−1	1.0568e−2	1.0794e +	1.0805e +	8.7976e−3	1.0383e +	**5.3625e−2**
	(5.81e−2) −	(1.72e−1) −	(2.13e−2) =	0 (1.41e−1) −	0 (2.21e−1) −	(1.91e−3) =	0 (1.90e−1) −	**(6.20e−3)**
+/−/=	3/8/1	2/9/1	0/8/4	0/12/0	0/9/3	1/7/4	0/12/0	

best results in 8 problems, demonstrating strong performance, followed by MMOEA-GPD, which secures the top results in 4 problems (Table 7).

For the IGDX evaluation, MMOEA-EGR also exhibits a comprehensive advantage, whereas MMOEA-GPD performs relatively poorly. The large standard deviation in its results indicates a lack of stability. Analyzing the population distributions obtained by the algorithms on the IDMPM2T2 test problem, as shown in Fig. 10, suggests that MMOEA-GPD often fails to identify a second equivalent Pareto solution set, contributing to its weaker performance in IGDX for this test suite.

Fig. 11 illustrates the final Pareto front obtained by each algorithm on the IDMPM3T2 problem. The results highlight that MMOEA-EGR excels in both convergence and diversity.

Based on the above analysis, the improved local convergence indicator I^i_{KLC} introduced in this paper is of great significance as a metric for the identification of derived individuals, which helps to enhance the performance of MMOEA-EGR in diverse multimodal multi-objective optimization problems. This strategy effectively enhances the adaptability of the algorithm in complex problem environments.

4.2.4. Analysis of the results of the HYL test suite

As an emerging evaluation standard for multimodal multi-objective optimization, the HYL benchmark specifically addresses problems exhibiting non-finite clusters of equivalent Pareto-optimal solutions. According to the literature (Ishibuchi et al., 2022a, 2022b), mainstream algorithms generally struggle with HYL test problems.

Tables 8 and 9 present the performance of the algorithms based on the IGD and IGDX metrics, respectively, where MMOEA-EGR achieves superior results on both.

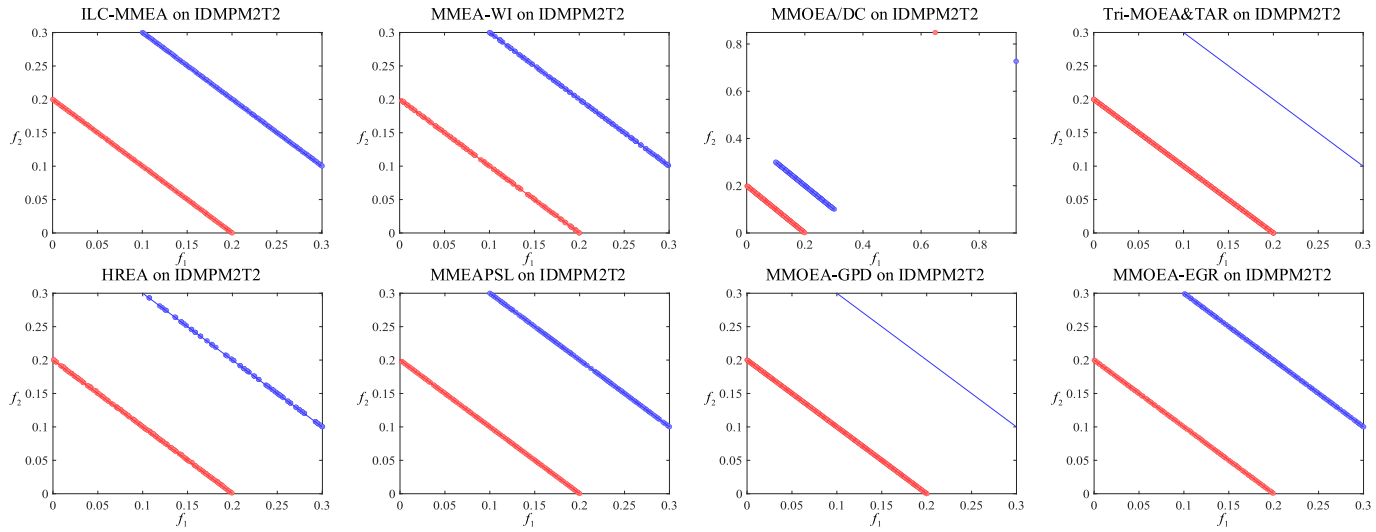


Fig. 10. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the IDMPM2T2 problem.

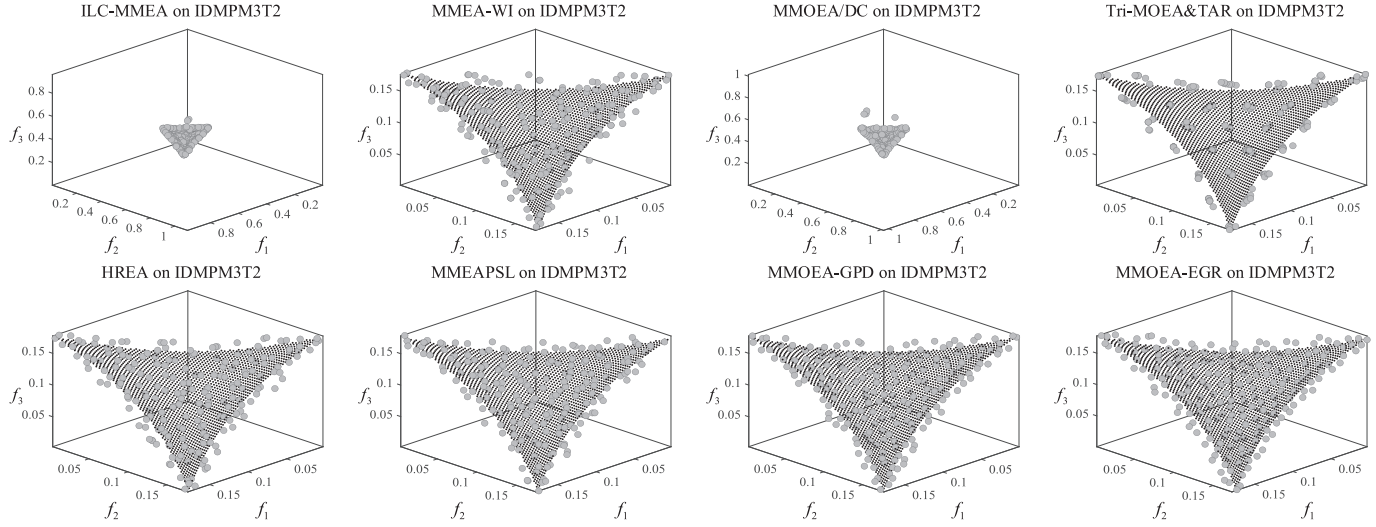


Fig. 11. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the IDMPM3T2 problem.

Table 8
Statistical results of IGD on HYL benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
HYL1	2.3125e +	2.3121e +	2.3120e +	2.3142e +	2.3122e +	2.2821e +	2.2418e +	**2.2280e +**
	0 (1.97e−4) −	0 (5.03e−4) −	0 (6.09e−4) −	0 (1.85e−4) −	0 (3.93e−4) −	0 (3.00e−4) −	0 (2.00e−4) −	**0 (6.12e−4)**
HYL2	7.0151e +	7.0160e +	7.0175e +	7.0150e +	7.0161e +	7.0165e +	7.0162e +	**7.0122e +**
	0 (1.02e−4) −	0 (8.64e−4) −	0 (1.50e−3) −	0 (8.07e−5) −	0 (1.02e−3) −	0 (1.44e−3) −	0 (1.07e−3) −	**0 (1.52e−3)**
HYL3	8.9907e +	9.0084e +	9.0230e +	**8.9893e +**	9.0007e +	9.0029e +	9.0123e +	9.0072e +
	0 (3.02e−3) +	0 (1.23e−2) =	0 (1.72e−2) −	**0 (3.16e−3) +**	0 (7.14e−3) +	0 (9.00e−3) =	0 (1.65e−2) =	0 (1.10e−2)
HYL4	8.9928e +	**8.9326e +**	8.9947e +	9.1128e +	9.0058e +	9.0168e +	8.9887e +	8.9888e +
	0 (1.04e−3) −	**0 (7.21e−4) +**	0 (1.34e−3) −	0 (3.92e−3) −	0 (3.28e−3) −	0 (5.26e−4) −	0 (1.25e−4) =	0 (1.34e−4)
HYL5	6.1354e +	6.1355e +	6.1356e +	6.1358e +	6.1355e +	6.1355e +	**6.1322e +**	6.1353e +
	0 (3.66e−4) −	0 (1.75e−4) −	0 (2.49e−4) −	0 (5.37e−5) −	0 (9.70e−5) −	0 (6.86e−5) −	**0 (4.91e−5) +**	0 (1.11e−4)
+/−/=	1/4/0	1/3/1	0/5/0	1/4/0	1/4/0	0/4/1	1/2/2	

Fig. 12 depicts the true Pareto solution sets (PS) for the HYL1 and HYL2 problems, while Fig. 13 illustrates the population distribution of the algorithms in the decision space for HYL1. The results indicate that most algorithms perform well in terms of population distribution, but MMOEA-EPD demonstrates stronger convergence, further emphasizing its stability and robustness in handling complex multi-objective optimization problems.

Additionally, Fig. 14 shows the population distribution in the decision space for HYL2. The analysis reveals that while the algorithms approach the optimal solution to some extent, MMOEA-EGR achieves a better balance between convergence and diversity, standing out in performance.

Overall, the HYL test suite further validates the superior performance of MMOEA-EGR in coping with complex MMOPs with infinite

Table 9
Statistical results of IGDX on HYL benchmark problems for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	Tri-MOEA&TAR	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
HYL1	2.0313e−1	2.0804e−1	2.0285e−1	3.5340e−1	1.9072e−1	1.9568e−1	1.8951e−1	**1.8712e−1**
	(8.47e−4) −	(3.16e−3) −	(1.67e−3) −	(8.18e−2) −	(1.39e−3) −	(1.91e−3) −	(2.06e−3) −	**(1.58e−3)**
HYL2	4.5346e−1	4.3965e−1	4.3632e−1	6.9669e−1	3.9244e−1	3.9623e−1	3.8973e−1	**3.8485e−1**
	(3.47e−3) −	(7.43e−3) −	(5.00e−3) −	(6.14e−2) −	(2.81e−3) −	(2.86e−3) −	(5.13e−3) −	**(2.09e−3)**
HYL3	9.7472e−1	6.6334e−1	7.0312e−1	7.5231e−1	6.3817e−1	6.2845e−1	**6.1214e−1**	6.2012e−1
	(2.17e−2) −	(9.71e−3) −	(1.19e−2) −	(5.28e−2) −	(5.45e−3) −	(5.09e−3) −	**(3.63e−3)** +	(7.77e−3)
HYL4	1.2598e +	9.5957e−1	1.0162e +	**8.8190e−1**	9.1952e−1	9.1546e−1	8.8388e−1	8.9933e−1
	0 (1.71e−2) −	(1.32e−2) −	0 (2.01e−2) −	**(6.59e−2)** +	(5.46e−3) −	(8.33e−3) −	(4.56e−3) +	(1.31e−2)
HYL5	1.9927e−1	1.9350e−1	2.3120e−1	2.4263e−1	1.7171e−1	1.7540e−1	1.7520e−1	**1.6885e−1**
	(1.71e−3) −	(4.06e−3) −	(5.34e−3) −	(1.21e−2) −	(1.93e−3) −	(1.46e−3) −	(1.32e−3) −	**(7.41e−4)**
+/−/=	0/5/0	0/5/0	0/5/0	1/4/0	0/5/0	0/5/0	2/3/0	

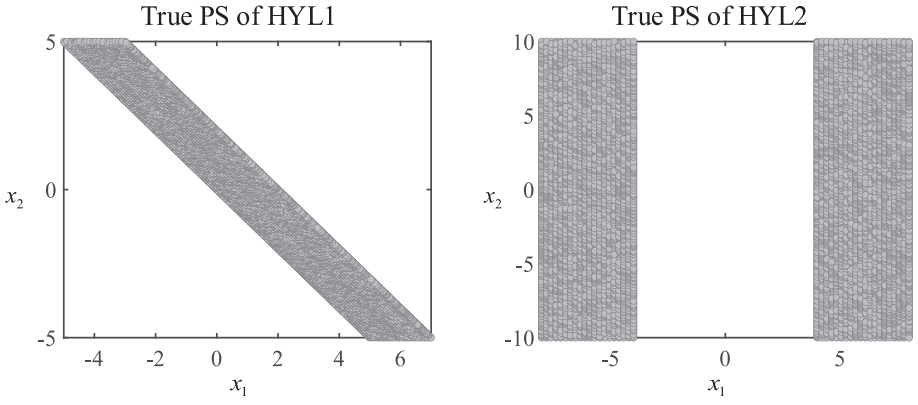


Fig. 12. True PS for HYL1 and HYL4 problems.

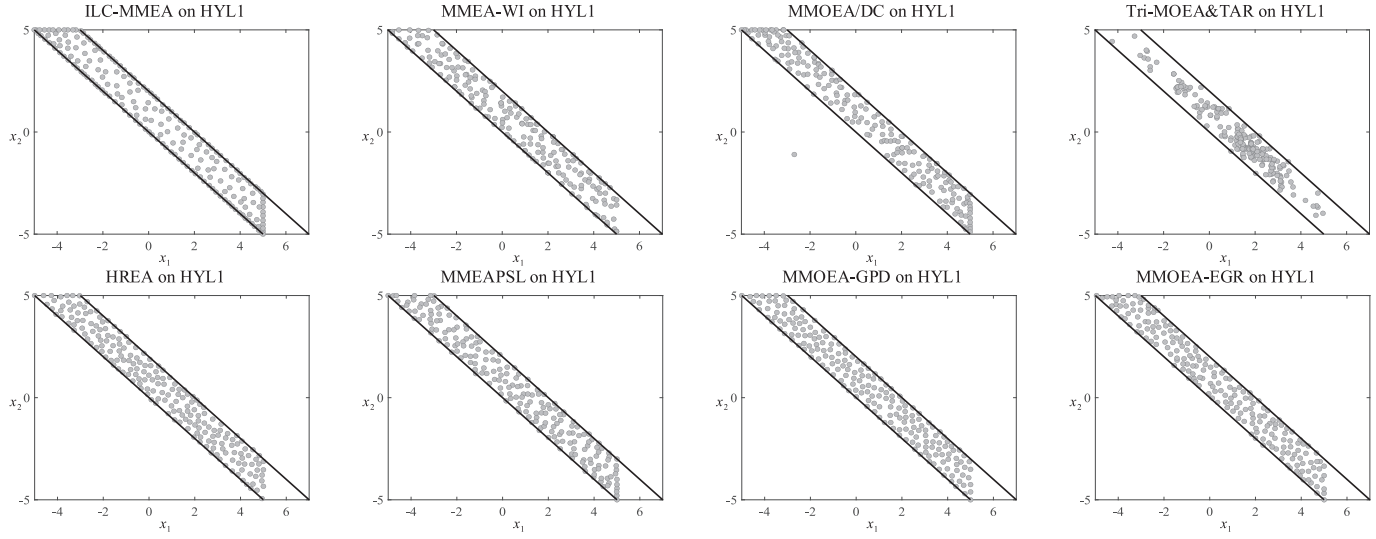


Fig. 13. The population distribution of all algorithms in the decision space for the HYL1 test problem.

equivalent PSs, showing the adaptability and versatility of its design.

4.3. Real-world multimodal multi-objective optimization problem

4.3.1. Real-world map-based distance minimization problem

The experiments in this section verify the effectiveness of the algorithms in dealing with the real-world map minimization problem (Ishibuchi et al., 2011). The problem is a two-dimensional four-objective multimodal multi-objective optimization problem with the objectives being the distances to the nearest elementary school, junior high school, convenience store, and subway entrance, respectively. The experimental settings are N = 100, E_{max} = 30000. The experimental results of each algorithm are shown in Figs. 15 and 16. The blue region represents the real Pareto solution set (PS) region provided by the authors of the literature, where elementary schools are represented by gray triangles, junior high schools by orange stars, convenience stores by green pentagrams, and subway entrances by purple squares, and the populations obtained by each algorithm are represented by red circles.

MMEA-WI in Fig. 15, and all algorithms in Fig. 16 show good convergence and diversity. In contrast, MMOEAPSL and MMOEA-EGR show significant advantages. Specifically, MMOEAPSL slightly outperforms MMOEA-EGR in the upper large PS region, while MMOEA-EGR

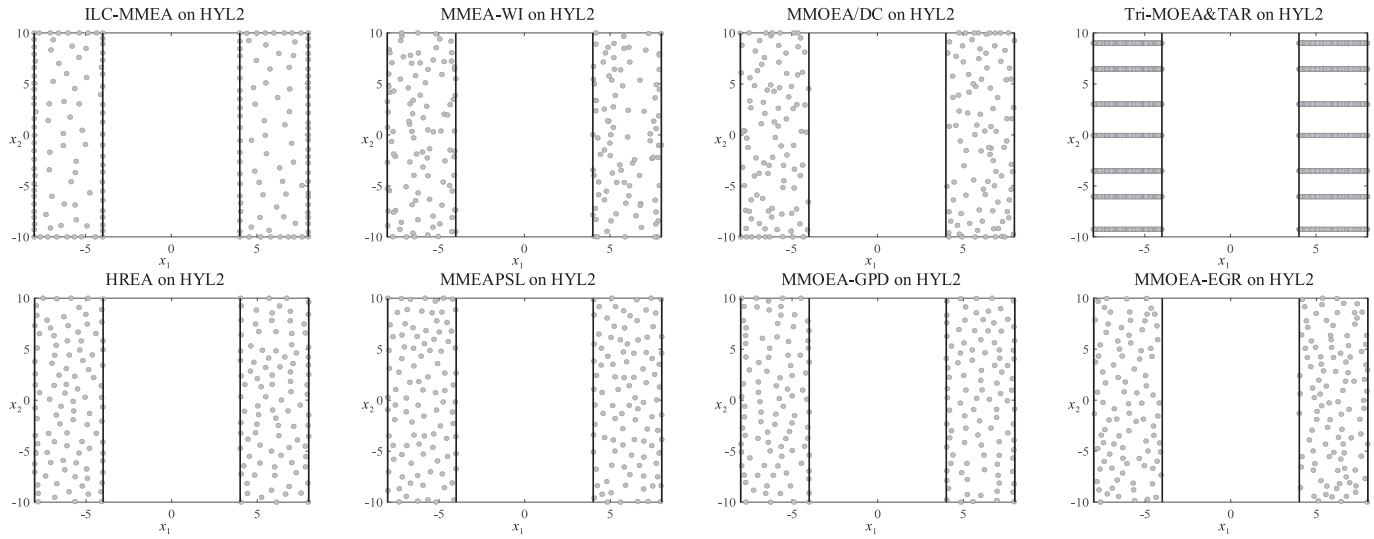


Fig. 14. The population distribution of all algorithms in the decision space for the HYL2 test problem.

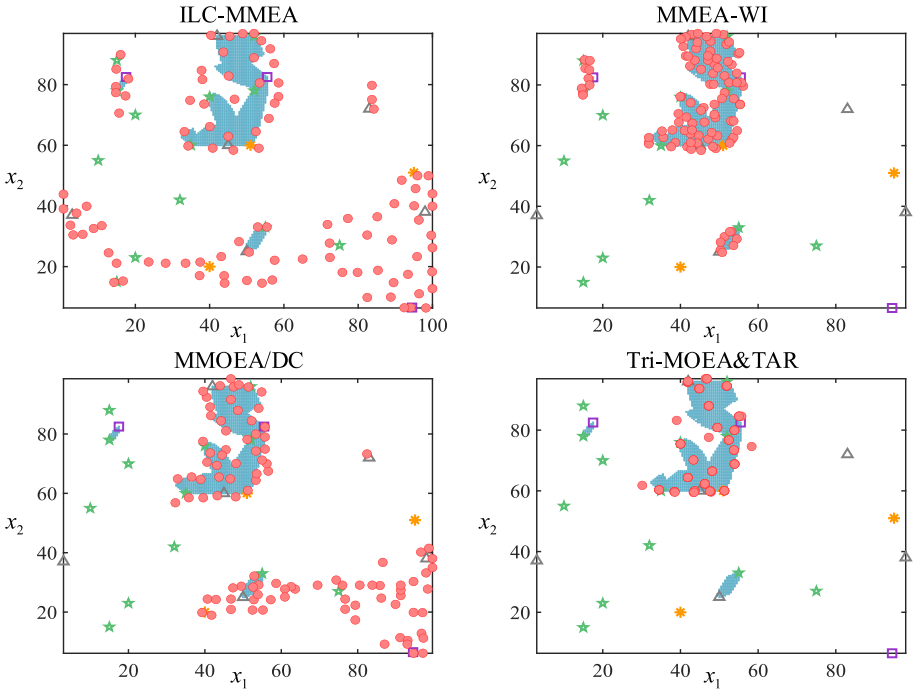


Fig. 15. Illustrates the population obtained by ILC-MMEA, MMEA-WI, MMOEA/DC and Tri-MOEA&TAR on this problem, as reflected in the decision space.

has a more balanced population distribution and performs better in the smaller PS regions below and to the upper left.

Overall, MMOEAPSL and MMOEA-EGR have their own focuses and are indistinguishable in the final results, which further validates the effectiveness of the two algorithms in real multimodal multi-objective optimization problems.

4.3.2. *Real-world multi-objective neural architecture search*

This section evaluates the performance of each of the compared algorithms in dealing with the real-world Multi-Objective Neural Architecture Search Problem (Ying et al., 2019) (MoNASP), which aims to optimize the structure of Convolutional Neural Networks (CNNs) with respect to two objectives: prediction error (f^e) and complexity based on the estimation of the number of weights (f^c). The problem is characterized by significant multimodality, which has been thoroughly explored in studies (Lu et al., 2023). In this paper, the CIFAR-10 dataset (Krizhevsky, 2009) is used to evaluate the generated CNN architectures, and the experimental parameters are set to a population size of N = 100 and a maximum number of function evaluations of 30,000. due to the lack of known true Pareto front (PF), this experiment uses the hypervolume metric (HV) as the performance measure, and each algorithm is run independently for 30 times.

The experimental results are shown in Table 10; the Tri-MOEA&TAR algorithm could not run successfully on this problem and therefore its results are not provided. In contrast, MMOEA-EGR achieved better performance than the other algorithms on this problem. The PFs obtained by each algorithm are shown in Fig. 17, where MMOEA-EGR exhibits good convergence and solution diversity, further validating its effectiveness in real applications.

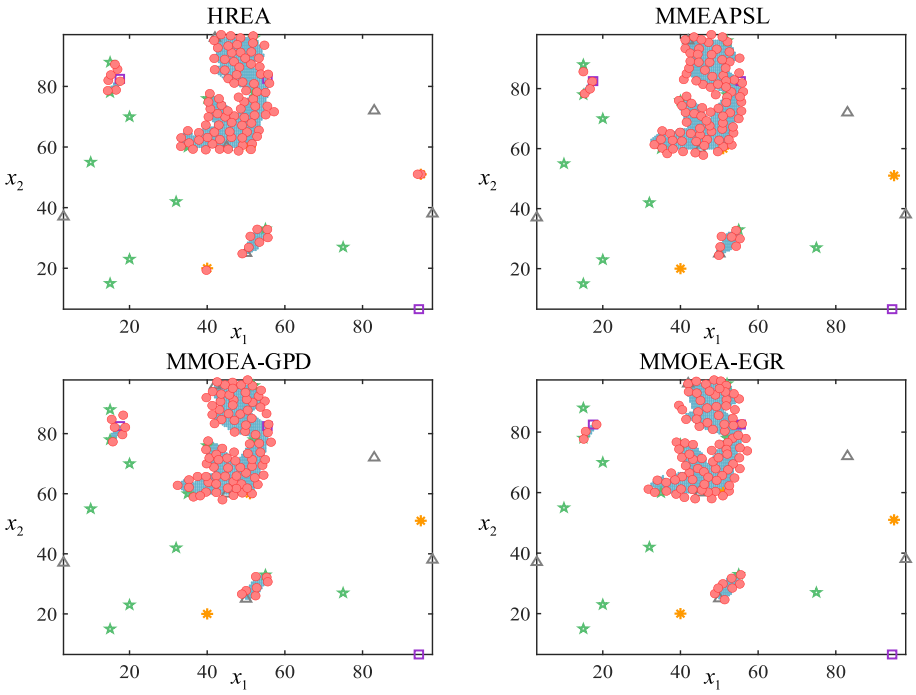


Fig. 16. Illustrates the population obtained by HREA, MMOEAPSL, MMOEA-GPD and MMOEA-EGR on this problem, as reflected in the decision space.

Table 10
Statistical results of HV on MoNASP problem for MMOEA-EGR and the comparison algorithms.

Problem	ILC-MMEA	MMEA-WI	MMOEA/DC	HREA	MMEAPSL	MMOEA-GPD	MMOEA-EGR
MoNASP	9.4926e−1 (1.05e−2) −	9.6952e−1 (8.95e−3) −	9.4751e−1 (9.29e−3) −	9.5025e−1 (1.02e−2) −	9.7256e−1 (4.56e−3) −	9.5868e−1 (5.82e−3) −	**9.7816e−1 (5.22e−3)**

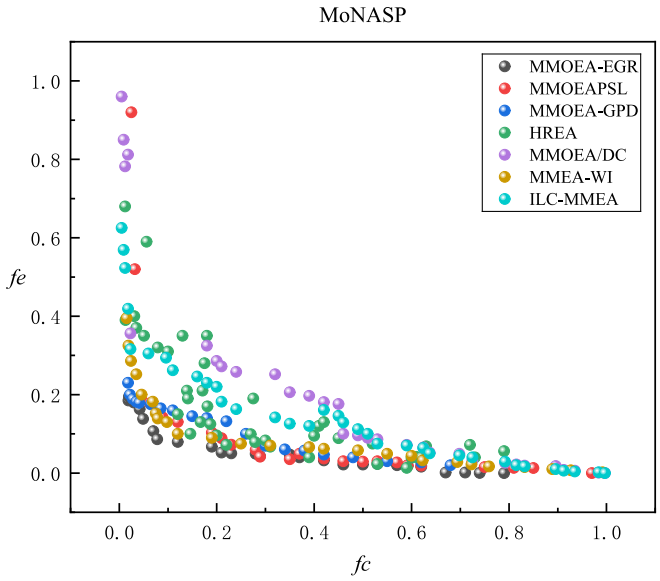


Fig. 17. The final Pareto front obtained by MMOEA-EGR and each comparison algorithm on the MoNASP problem.

4.4. Ablation experiment

To further explore the MMOEA-EGR algorithm, this study investigates the specific roles of the three proposed improvement strategies and their contributions to the overall optimization process. To achieve this, the following comparative experiments were designed and implemented. Specifically, three algorithm variants were constructed based on MMOEA-EGR, each omitting one of the improvement strategies to assess its impact:

- MMOEA-EGR_Variant1: This variant removes the reinforcement learning-based evolutionary stage selection to evaluate the generalizability of this strategy across diverse MMOPs.
- MMOEA-EGR_Variant2: This variant of the algorithm omits the remodeling of the distribution of derived populations strategy, in order to explore the role of this strategy in improving the overall quality of derived populations and promoting evolutionary efficiency.
- MMOEA-EGR_Variant3: This variant replaces the derived indicator with the original MMOEA-GPD indicator to analyze the generalizability of the improved local convergence indicator in identifying promising solutions across different MMOPs.

A comprehensive performance evaluation of MMOEA-EGR and its three variants was conducted on representative test suites. The experimental results, presented in Tables 11 and 12, clearly demonstrate the significance of each improvement strategy.

Specifically, by comparing MMOEA-EGR_Variant1 and MMOEA-EGR, it can be found that the selection of the three evolutionary stages using reinforcement learning when dealing with diverse MMOPs can dynamically match the most suitable evolutionary stage for each iteration throughout the evolutionary process, thus better balancing the diversity of the population with the convergence. By comparing the experimental results of MMOEA-EGR_Variant2 and MMOEA-EGR, it can be seen that the strategy of remodeling the distribution of derived populations significantly improves the optimization efficiency, which enables the algorithm to perform better in both IGD and IGDX metrics with a fixed number of iterations as the termination condition. In addition, by comparing MMOEA-EGR_Variant3 with MMOEA-EGR, it

Table 11
Statistical results of MMOEA-GPD and each algorithm variant on the IGD metric for the test functions, with the best results in each row highlighted.

Problem	MMOEA-EGR_Variant1	MMOEA-EGR_Variant2	MMOEA-EGR_Variant3	MMOEA-EGR
MMF1	3.2931e−3 (3.07e−4) −	3.0724e−3 (2.16e−4) =	3.1025e−3 (2.24e−4) =	**3.0688e−3 (2.65e−4)**
MMF2	1.3926e−2 (2.02e−3) −	**1.0682e−2 (1.05e−3)** =	1.4124e−2 (1.53e−3) −	1.0703e−2 (9.61e−4)
MMF3	1.2621e−2 (2.38e−3) −	9.8206e−2 (1.12e−3) =	**9.7984e−3 (8.26e−4)** =	9.8946e−3 (7.02e−4)
MMF4	**2.6397e−3 (3.12e−4)** +	2.9197e−3 (2.52e−4) =	2.8124e−3 (5.23e−4) =	2.9309e−3 (3.15e−4)
MMF5	3.2015e−3 (2.67e−4) =	3.1225e−3 (2.55e−4) =	3.2128e−3 (2.73e−4) =	**3.1026e−3 (2.46e−4)**
MMF6	3.0251e−3 (4.16e−4) =	3.0096e−3 (3.74e−4) =	3.0286e−3 (5.22e−4) =	**3.0004e−3 (2.17e−4)**
MMF7	2.7422e−3 (2.33e−4) =	2.7524e−3 (2.61e−4) =	**2.7386e−3 (2.96e−4)** =	**2.7988e−3 (2.67e−4)**
MMF8	2.2025e−3 (4.27e−4) =	**1.8224e−3 (2.07e−4)** +	2.4924e−3 (1.58e−4) −	2.1987e−3 (5.84e−4)
MMMOP1_A	1.2924e−3 (4.82e−5) −	1.2519e−3 (3.26e−5) −	1.0205e−3 (5.72e−5) =	**1.0155e−3 (4.93e−5)**
MMMOP2_B	3.1726e−2 (2.35e−3) =	3.1632e−2 (5.19e−3) =	**3.1523e−2 (2.15e−3)** =	3.1609e−2 (4.76e−3)
MMMOP3_A	**1.2328e−3 (3.05e−5)** +	1.6055e−3 (3.81e−5) =	1.3102e−3 (1.79e−5) +	1.6667e−3 (3.43e−4)
MMMOP4_B	**1.0728e−1 (7.62e−2)** =	1.0927e−1 (5.35e−2) =	1.3612e−1 (7.92e−2) −	1.0850e−1 (8.84e−2)
MMMOP5_C	2.5504e−3 (1.02e−3) =	2.5526e−3 (2.91e−3) =	1.0519e−2 (2.32e−2) −	**2.5462e−3 (9.38e−4)**
MMMOP6_D	3.6012e−2 (2.90e−3) −	3.5936e−2 (3.82e−3) −	3.5826e−2 (2.36e−3) −	**3.3982e−2 (2.61e−3)**
IDMPM2T2	4.5082e−4 (3.13e−5) =	**4.4265e−4 (1.97e−5)** =	4.5205e−4 (2.23e−5) =	4.4988e−4 (2.97e−5)
IDMPM3T3	5.3901e−3 (1.52e−4) −	5.4025e−3 (2.47e−4) −	5.3815e−3 (1.72e−4) −	**4.9858e−3 (1.80e−4)**
IDMPM4T4	6.7908e−3 (4.05e−4) =	**6.7759e−3 (5.71e−4)** =	6.8021e−3 (3.26e−4) =	6.7812e−3 (3.24e−4)
HYL1	2.2527e + 0 (2.32e−4) −	2.2452e + 0 (4.84e−4) −	2.2403e + 0 (2.57e−4) −	**2.2280e + 0 (6.12e−4)**
HYL2	7.0163e + 0 (1.12e−3) −	7.0159e + 0 (1.56e−3) −	7.0130e + 0 (1.47e−3) =	**7.0122e + 0 (1.52e−3)**
HYL3	**8.9795e + 0 (2.05e−2)** +	9.0085e + 0 (4.75e−2) =	9.0102e + 0 (1.29e−2) =	9.0072e + 0 (1.10e−2)
HYL4	**8.9422e + 0 (2.10e−4)** +	8.9852e + 0 (3.94e−4) =	8.9889e + 0 (1.28e−4) =	8.9888e + 0 (1.34e−4)
HYL5	6.1352e + 0 (9.12e−5) =	6.1355e + 0 (1.29e−4) =	**6.1334e + 0 (1.06e−4)** +	6.1353e + 0 (1.11e−4)
+/−/=	4/8/10	1/5/16	2/7/13	

Table 12
Statistical results of MMOEA-GPD and each algorithm variant on the IGDX metric for the test functions, with the best results in each row highlighted.

Problem	MMOEA-EGR_Variant1	MMOEA-EGR_Variant2	MMOEA-EGR_Variant3	MMOEA-EGR
MMF1	**4.3527e−2 (2.58e−3)** +	4.9525e−2 (5.01e−3) −	4.8089e−2 (3.74e−3) =	4.7405e−2 (4.92e−3)
MMF2	3.2292e−2 (5.21e−3) −	1.8815e−2 (3.12e−3) =	2.0242e−2 (3.98e−3) −	**1.8748e−2 (2.73e−3)**
MMF3	3.0926e−2 (3.17e−3) =	3.1052e−2 (4.24e−3) =	3.0986e−2 (3.81e−3) =	**3.0856e−2 (1.48e−2)**
MMF4	2.0027e−2 (2.42e−3) −	1.3682e−2 (2.08e−3) =	1.3617e−2 (2.94e−3) =	**1.3501e−2 (2.48e−3)**
MMF5	6.8159e−2 (4.52e−3) −	6.6407e−2 (6.24e−3) −	5.9319e−2 (5.57e−3) =	**5.9233e−2 (6.56e−3)**
MMF6	**6.0514e−2 (4.83e−3)** +	6.5809e−2 (3.15e−3) =	6.5732e−2 (3.26e−3) =	6.5686e−2 (3.45e−3)
MMF7	1.7654e−2 (8.32e−4) −	1.7489e−2 (3.12e−3) −	1.7831e−2 (6.21e−3) −	**1.4450e−2 (2.53e−3)**
MMF8	6.9310e−2 (6.64e−2) −	3.9012e−2 (4.15e−3) =	**3.8801e−2 (4.60e−3)** =	3.8953e−2 (5.00e−3)
MMMOP1_A	5.0179e−3 (2.56e−2) −	1.9713e−3 (1.06e−3) =	**1.5207e−3 (3.15e−3)** +	2.0935e−3 (8.99e−4)
MMMOP2_B	3.5104e−2 (2.79e−3) +	**3.4607e−2 (2.01e−3)** +	4.7903e−2 (9.75e−3) =	4.7857e−2 (8.01e−3)
MMMOP3_A	4.1913e−3 (2.15e−3) =	5.2849e−3 (3.70e−3) −	5.5814e−3 (4.27e−3) −	**4.1866e−3 (2.39e−4)**
MMMOP4_B	1.1276e−1 (6.27e−2) −	8.7218e−1 (4.51e−2) −	8.9652e−2 (8.92e−2) −	**8.2098e−2 (5.13e−2)**
MMMOP5_C	**8.8943e−3 (2.06e−3)** +	5.0192e−2 (5.16e−2) +	5.8904e−2 (7.51e−2) =	5.8391e−2 (8.84e−2)
MMMOP6_D	**1.1314e−1 (4.91e−3)** +	1.9512e−1 (6.24e−2) −	1.7104e−1 (5.27e−2) =	1.7182e−1 (7.10e−2)
IDMPM2T2	6.6809e−4 (4.10e−5) =	**6.5218e−4 (5.28e−5)** =	5.2606e−1 (4.12e−1) −	6.6132e−4 (4.36e−5)
IDMPM3T3	8.1804e−3 (7.01e−4) =	8.1924e−3 (6.12e−4) =	2.5125e−1 (5.13e−1) −	**8.1795e−3 (5.55e−4)**
IDMPM4T4	5.3801e−2 (5.14e−3) =	5.3772e−2 (4.19e−3) =	3.2813e−1 (1.71e−1) −	**5.3625e−2 (6.20e−3)**
HYL1	1.9014e−1 (3.36e−3) −	1.8716e−1 (2.24e−3) =	1.8730e−1 (1.40e−3) =	**1.8712e−1 (1.58e−3)**
HYL2	3.8817e−1 (4.21e−3) −	3.8490e−1 (7.10e−3) =	3.8917e−1 (5.12e−3) −	**3.8485e−1 (2.09e−3)**
HYL3	**6.1222e−1 (4.13e−3)** +	6.3527e−1 (4.15e−3) −	6.3258e−1 (3.24e−3) −	6.2012e−1 (7.77e−3)
HYL4	8.8380e−1 (4.21e−3) +	**8.8302e−1 (7.10e−3)** +	8.9956e−1 (2.96e−2) =	8.9933e−1 (1.31e−2)
HYL5	1.8204e−1 (4.75e−3) −	1.8901e−1 (2.14e−3) −	1.7981e−1 (5.67e−3) −	**1.6885e−1 (7.41e−4)**
+/−/=	7/10/5	3/8/11	1/10/11	

can be seen that the improved local convergence indicator enables MMOEA-EGR to more accurately identify solutions with exploratory potentials, thus helping the algorithm to discover more equivalent Pareto subsets.

4.5. MMOEA-EGR computational complexity analysis

In each generation, the computational cost of the MMOEA-EGR algorithm primarily stems from its three evolutionary stages and the reinforcement learning mechanism. Specifically, in the convergence stage, the algorithm performs crossover and mutation operations ($O(N)$) on a population of size N, followed by fitness evaluation and truncation selection ($O(N^2) + O(N\log N) \approx O(N^2)$). During the population derivation stage, new individuals are generated by repeatedly evaluating the fitness of both the original and derived populations ($O(N^2)$) and applying truncation selection ($O(N\log N)$), with the dominant cost denoted as ($O(N^2)$). The diversity maintenance stage similarly requires fitness evaluations and environmental selection on the collaborative population, with a computational complexity denoted as ($O(N^2)$).

For the reinforcement learning component, the decision-making process involves computing the mean value of N individuals in a D-dimensional decision space to construct the state vector ($O(ND)$)), as well as training the neural network, where the training complexity ($O(D^2)$) is the main source of overhead. Other operations, such as reward calculation and experience replay, typically have linear or constant computational complexity.

Taken together, the overall computational complexity of MMOEA-EGR per generation can be expressed as $O(N(N+D)+D^2)$, where N is the population size and D is the decision variable dimension.

Although the introduction of the reinforcement learning mechanism increases the computational complexity of MMOEA-EGR compared to traditional MMOEA, the performance improvement it brings is significantly better than this cost. In multimodal multi-objective optimization tasks, MMOEA-EGR exhibits enhanced optimization capabilities. This computational overhead is fully affordable given the continuous advances in modern computing hardware.

5. Conclusions and future work

This study introduces a novel multimodal multi-objective

evolutionary approach MMOEA-EGR, which enhances goal orientation and restructures population distributions. MMOEA-EGR dynamically selects different evolutionary stages using deep neural networks, ensuring efficient collaboration throughout the optimization process. In the convergence stage, the algorithm accelerates the population's approach to the Pareto front, improving convergence speed. In the population derivation stage, it actively searches for individuals beneficial to decision space exploration, optimizes computational resource allocation within sub-regions, and enhances both diversity and evolutionary efficiency by restructuring population distribution. In the diversity maintenance stage, the algorithm ensures an even distribution of individuals in both decision and objective spaces, balancing convergence and diversity. These three stages work in synergy to improve decision space searchability and significantly enhance population performance in terms of both convergence and diversity.

Future work will extend MMOEA-EGR to high-dimensional and constrained MMOPs. Key challenges include maintaining robustness in expanded decision spaces and handling constraints effectively, especially for real-world applications demanding both strict feasibility and multimodal solutions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Originality statement

We confirm that the work presented in this manuscript is original and has not been published elsewhere, and that it is not under consideration for publication elsewhere, in whole or in part.

Data availability

Data will be made available on request.

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